Big Data Analysis in Quality Control of Manufacturing SMEs

Session 2021-2025

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Undertaking

We hereby certify that the research work titled "Big Data Analysis in Quality Control of Manufacturing SMEs" is our original creation. The work has not been submitted for evaluation to any other institution. Any content derived from external sources has been appropriately acknowledged and cited.

We have carefully acknowledged all sources and references used during our research. We also confirm that we have not used any Generative AI tools (such as ChatGPT or similar) to write, analyze, or generate any part of this project.

If at any point in the future it is found that AI tools were used in violation of this statement, we accept that we will be held accountable.

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Abstract

The project examines the integration of a technology-oriented, data-based model to improve SME manufacturing quality. By considering the limitations of having limited resources, not sufficiently managing the defect, and the lack of predictive tools, the framework aims to support the quality of products and operational efficiency.

A powerful algorithmic approach has been established that embedded machine learning in combination with data-mining and causal analytic procedures such as classification models, association rule mining, causal discovery, and counterfactual simulation. The process of the research involved gathering data, preprocessing, training model and extracting patterns. The multilayer perceptron (MLP) model was used to sort the defects based on the severity and with the highest precision, predict whether a rework was required. The association rule mining methodology showed common relationships between triggers of defects and types of defects, and causal discovery unveiled the underlying causes of problems with quality.

The counterfactual analysis built an estimate of likely outcomes of possible interventions, thus providing specific data of process improvement. In general, the project helps to develop intelligent manufacturing by allowing the predictive quality control, reducing rework, and improving data-based decision-making. It provides operational sustainability and long-term competitiveness, too, as well as small and middle-sized manufacturers.

Acknowledgement

We would like to express our deepest gratitude to our supervisor, Dr. Abid Ali, for his invaluable guidance, unwavering support, and expert advice throughout the entire duration of this project. His insightful feedback and encouragement were instrumental in shaping our research and pushing us to achieve our best.

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Chapter 1: Introduction

In this chapter, an outline of the structure and content of the thesis will be provided. It introduces the subject of the research, defines the purposes, explains the methodological practices, highlights the main uses and lists the benefits and drawbacks of the study, thus referring to the Sustainable Development Goals (SDGs) as shown in Flow chart Chapter 1

Figure 1.

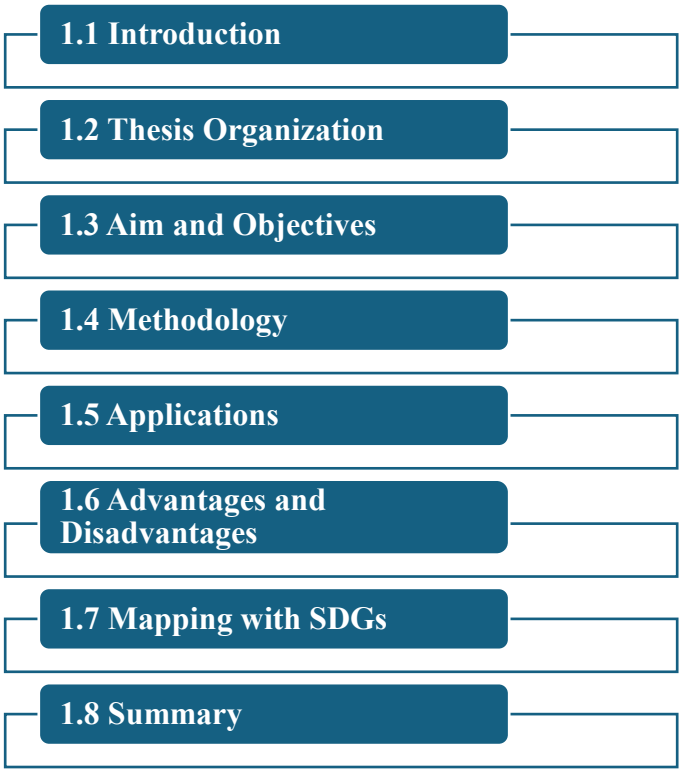


Figure 1 Flow chart Chapter 1

1.1 Introduction

The capacity to make sound judgments in the era described by the high rates of technological advancement and data-driven innovation is highly decisive in terms of success in different industries. The current project provides a cutting-edge Python-based framework that combines Association Rule Mining, Causal Inference, and Counterfactual Analysis and translates raw data into action. The framework goes beyond the traditional analytical processes because it reveals hidden patterns, identifies the cause of the complex problems, evaluates the future

possibilities to guide strategic decision-making. Using the immense potential of big data and machine learning, the solution contends with the most significant issues such as the promotion of quality management, minimizing structural waste of operations, and increasing the general efficiency. The proposed framework is developed with a view to a competitive, fast-developing, quality-oriented enterprise and equips organizations with the operational tools that are needed at the time and in the future to support their evidence-based decisions and to promote sustainable organizational growth and maintain innovation. Therefore, the project does not only become the means of helping business adapt to the changing market conditions but also ensures the business long-term success in a more data-driven world.

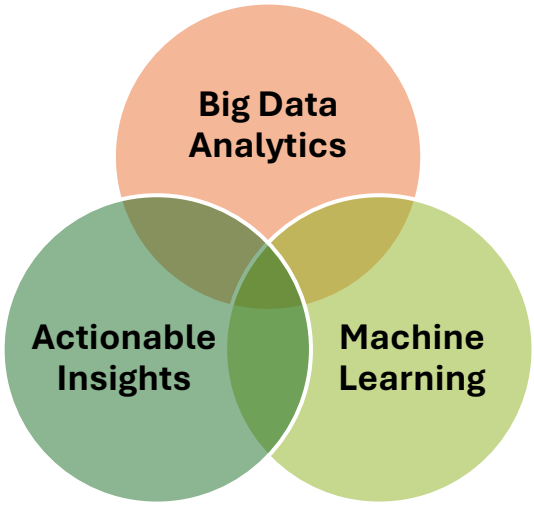


Figure 2 Project Synergy Diagram

1.1.1 Problem Statement

The critical issue that loses many Small and Medium-Sized Enterprises (SMEs) the ability to maintain a consistent quality of product is attributed to poor quality management systems based on reactive and out dated practices that is incapable of meeting the demands of an ever competitive and globalized markets. These inefficiencies result in overproduction, increased cost of rework, and lack of competitiveness hence hindering the capacity of their sustainable growth and responsiveness to the changing customer demands. The current project aims at addressing such pivotal dilemmas by building an assertive, data-driven quality management model facilitated by big data analytics and machine learning. The framework improves defect identification, rework and resource efficiency through application of high level techniques. As a result of this, SMEs have all the tools to move into predictive quality management reshaping

operations excellence, supporting sustained growth, and achieving competitive advantage in an environment of fast changes.

1.2 Thesis Organization

This current study is a systematic inquiry into the level to which highly sophisticated algorithms and data-driven solutions could improve the manufacturing process by perfecting the quality control inefficiencies. The key target is to realign the traditional quality-management procedures with data-driven processes that fine-find defect identification, reduce operational redundancies, and instill sustainable processes. This project is based on a well-structured eight-phase approach to methodology, described in Chapter 3, which provides the investigators with a clear working route through the complex process of analysis and optimization. Every step that has to be taken, starting with the data collection and up to the results assessment, has a specific (but, at the same time, interrelated) purpose, and as a result, a complete view of what makes a product high quality and the completed operations more efficient is achieved. Initial phases of the model are focused on a data infrastructure set-up. The process of Data Collection collects complex data that includes metrics on the process and defect measurements and other variables of operation. Such aggregation occurs during the Data Cleaning and Pre-processing stages during which inconsistencies are eliminated and the data set is organized to form future analysis. The following stages move away to the creation of information towards the information extraction. During the Defect Classification phase, Neural-network model (a multi-layer perceptron) will be used in order to obtain accurate detection of faulty objects. In the Pattern Discovery stage, the obvious associations are to be mined to reveal clear links between manufacturing parameters and emergence of defects. Root Cause Analysis applies causal inference in order to go beyond correlation, stating causes explicitly in terms of effects. The Scenario Evaluation stage allows the evaluation of counterfactual analyses that would allow alternatives in action to be simulated which provides predictive judgments on alternative actions. The final Phase will combine these results into a more user-friendly structure that bundles it into a result package using various visualizations and other interactive tools. Such an arrangement provides the stakeholders with the ability to make evidence-based decisions, look into the quality challenges, and streamline the production policies. Machine-learning algorithms combined with statistical inference do not only offer the level of rigor needed by

scientific research, but also provide the flexibility of use that contemporary industry requires. Taken together, the current study constitutes an alliance in which innovative algorithms, powerful data analyses, and practical execution approaches collide. Maximizing the potential of big data and machine learning, the research will enhance the understanding of the dynamics of quality management and, at the same time, provide industries with life-changing tools to increase productivity, reduce waste, and promote sustainable development.

1.3 Aim and Objectives

1.3.1 Aim

Machine-learning and data-analytics enabled hybrid Python-based platform will transform the process of quality management, allowing its elements, including defect analysis, root-cause determination, informed and foreseeing decision-making, to offer even greater efficiency to operations and become the source of sustained growth in a manufacturing context.

1.3.2 Objectives

i Establish an analytical approach to data collection, cleaning, pre-processing that will consist in securing reliability and ensuring the data is ready to be analyzed further.

ii By use of Association Rule Mining, one is able to detect patterns, as well as to correlate defects.

iii Causal inference can be applied in the establishment and confirmation of the cause of defects.

iv Carry out a counterfactual analysis to rate different strategies and predict their impact on the indicators of quality.

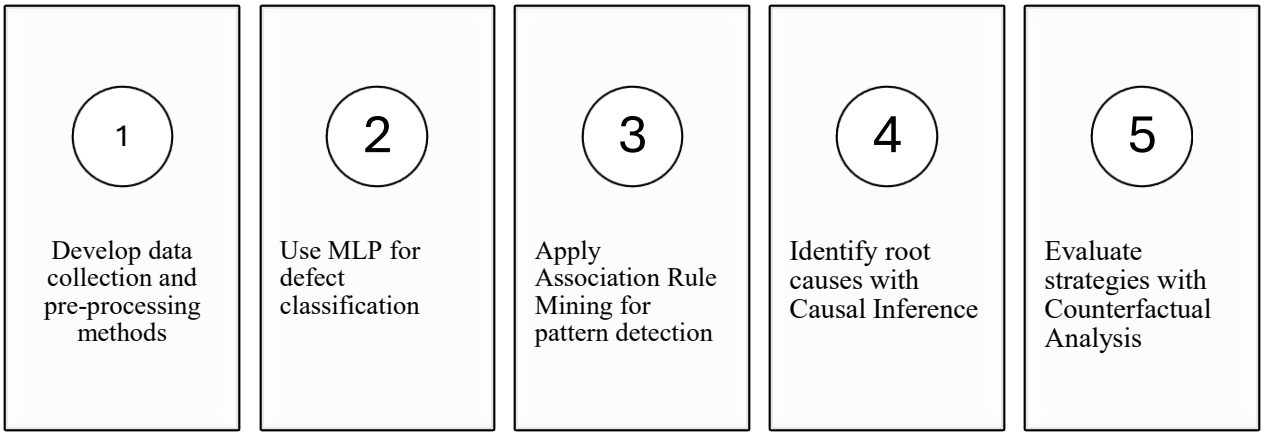


Figure 2 Project Objectives

1.4 Methodology

The section presents a highly-developed methodology that aims at transforming the quality management within manufacturing environments by means of using advanced, data-driven solutions. The suggested framework consists of five interconnected processes, including preprocessing the data, defect categorization, pattern finding, root cause examination, and counterfactual analysis aimed to improve the level of quality control and operational productivity. The first phase is assembly and pre-treatment of the data. Various datasets such as the product specifications, defects, production, and operational conditions are derived can occur in the information resource involving credible and reliable sources of information. Through strict cleansing, and transformation, these datasets eliminate inconsistencies, outlier, and missing elements, increasing the accuracy of the analysis and data readiness. The next step is devoted to the classification of defects. The Multi-Layer Perceptron (MLP) method of machine learning could be used to detect and classify defective goods. The ensuing categorization details the trends in defects and offers detailed information on severity of defects, which form a basis of further analysis and intervention into the same. Then a pattern is discovered, using association rule mining (ARM) -based methods (in particular, Apriori algorithm) in order to find the ties and associations between manufacturing factors and defect occurrence. This step provides executable patterns such as the interaction of variables that generate defects hence making superior decision making and optimization of the process. The root cause analysis and counterfactual evaluation are then carried out using Propensity Score Matching (PSM) framework of robust causal discovery. In the root cause analysis stage of the model, LINGAM is used in determining causal correlation between production variables and defects beyond the simple correlation to provide the underlying cause of quality problems. The counterfactual analysis also uses the PSM and models alternative postconditions that forecast the implications behind various strategic decisions. This will give the stakeholders insight on possible ways to be hit and make decisions based on data to reduce possible defects and achieve maximum quality results. This combination of the five phases: data preprocessing, defects classification, pattern discovery, and advanced causal and counterfactual analysis makes this methodology a solid platform of quality improvement. It ensures accurate identification of defects, profound understanding on cause and effect relations, as well as forecasting of strategic actions. The end-to-end solution is not only likely to lessen the rate of defects and operational efficiencies, but it

also spurs sustainable growth and enables industries to be competitive in an ever-changing playing field.

1.5 Applications

The current approach provides a data-driven and structured system aimed at enhancing quality management of manufacturing businesses and therefore facilitating proactive decision-making and increased operational efficiency. With the application of advanced analytics and machine-learning method, this framework can easily be implemented in many industrial sectors:

1. Diminishing defect frequencies through correct grouping and deep root-cause diagnosis.

2. Find the best way to allocate resources and reduce wastage in the course of the production.

3. Improving the performance of the supply chain through identification of important quality-influencing variables.

4. The ability to predict the impact of process changes to influence decision-making activities.

5. Enhancement of sustainability by minimizing inefficiency of materials and energy.

1.6 Advantages and Disadvantages

1.6.1 Advantages

Enhanced Quality Control:

Implementation of machine learning model in the SMEs allows business organizations to predict the nature of quality defects and prevent their future manifestations to reduce variability in production and minimize the amount of good products going off specifications. Such preventive indication maintains the quality of manufacturing which also increases customer satisfaction.

Efficiencies Cost:

SMEs minimize rework, wastage and overall production spending by identifying shortcomings in the early stages and efficient distribution of resources. To companies that expect the

deterioration of quality, this strategy will allow avoiding excessive defence expenditures as opposed to making the best out of the wrong that has already happened.

Data-Driven Decision Making:

The models give tangible advice based on real-life data that management could develop to improve quality and increase productivity in the organization. As a result, the resources which are available scarce can then be directed to where they will have the greatest returns.

Scalability and Flexibility:

The project is based on such machine learning techniques that allow the SMEs to apply the technique to other lines of production or even allow such an application to provide expansion without significant alteration. Since every MSME application is highly configurable, then it can fit any specific operational needs.

Skill development of the Employee:

Involvement in the realization of such an advanced technology and data-driven decision-making system makes the employee more competent and builds a culture of continuous development, or a culture of self-administration called Kaizen.

1.6.2 Disadvantages

- **Initial Setup Costs**:

  Incorporating a new technological breakthrough in the area of machine-learning solutions requires a high amount of initial capital investment to obtain the necessary technology, software and human resources. In case of small- and medium-sized enterprises (SMEs), this spending at the deployment stage can be an unsurmountable obstacle, as their financial potential is rather limited.

- **Data Quality Requirements**:

  Any predictive model is itself dependent on the quality of data used in the training of that model. Therefore, such cases of imprecision, incompleteness or inconsistency in such data may lead to the prediction with doubtful reliability, and, therefore, the effectiveness of the subsequent quality management scheme will be impaired.

- **Complexity of Implementation**:

  The effectiveness of a model depends on the quality of data to which they are trained to; hence these models can be prone to falsehoods, gaps, inconsistency, all of which can affect predicted reliability harming the overall usability of a quality management system.

- **Dependence on Technology**:

  The usage of machine learning modeling will also place on small-to-medium-sized enterprises the obligation to acquire the necessary infrastructure including enough space to store the data, the necessary computing capabilities, and the software maintenance needs that such systems require. Any disruption of these components may jeopardize production quality.

- **Resistance to Change**:

  Managements and employees might not react positively when new technologies are brought on board especially when practices of quality-management have been there in the operations over a long period of time. Such innovation should be properly implemented through effective change management and specific training, which are thus invaluable. Notably, despite the significant benefits of machine learning to small and medium-sized enterprises (SMEs), such as the enhancement of product quality and the increase in efficiency, this technology has several drawbacks, which are the high initial investments, strict data prerequisites, as well as the complex nature of implementation. However, should SMEs meet such stumbling blocks by taking time in planning and investing suitably, the benefits soon overshadow the disadvantages and can support the growth of such firms through a long term perspective without losing competitiveness.

1.7 Mapping with SDGs

1.7.1 SDG 8: Decent Work and Economic Growth

Our project is dedicated to stimulating the long-term, universal, and sustainable financial growth by supporting small and middle-sized companies (SMEs) to increase production rates and efficiency of the operation. Having reduced flaws, decreased the amount of waste products, SMEs will be able to achieve a higher level of productivity and create more respectable job opportunities, therefore, contributing to the overall growth of the economy. Besides, through the use of new technologies, SMEs manage to remain competitive, keeping the stable employment rates and improving the working conditions among employees.

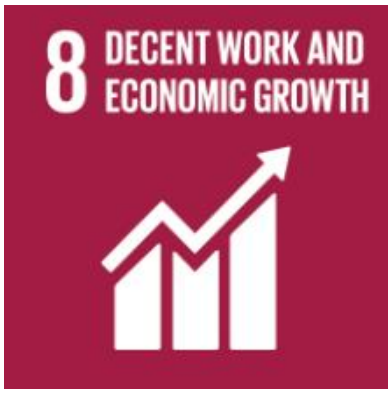


Figure 3 SDG 8Decent Work and Economic Growth

1.7.2 SDG 9: Industry, Innovation, and Infrastructure

The use of the machine learning approaches and data-driven decision-making processes is also based on SDG 9 because it leads to innovations in the manufacturing industry. The current project influences SMEs to find the opportunity to use advanced technologies, thus, contributing to the sustainable industrialisation and resilient infrastructure. The initiative helps in establishing a sustainable system of production that can develop stronger production systems by allowing SMEs to adopt innovative production means which in turn enhance further growth in the industry as well as bringing about more industrialization and modernization.



Figure 4 SDG 9: Industry, Innovation, and Infrastructure

1.7.3 SDG 12: Responsible Consumption and Production

This project contributes to Sustainable Development Goal 12 by reducing the amount of rework, eliminating the excessive generation of inefficient resources, and waste. Through small and medium-sized enterprises embracing the practice of proactive quality management, they simply promote responsible consumption and production hence reducing the impact of manufacturing activities on the environment. This is meant to encourage a resource-conserving manufacturing process, the key to the long-term sustainability of manufacturing business, such that they are, at the same time, efficient and ecological.



Figure 5 SDG 12: Responsible Consumption and Production

1.7.4 SDG 13: Climate Action

The elimination of waste and maximization of resource utilization, a strategy that the project applies, contributes towards the United Nations Sustainable Development Goal 13, which is about climbing the challenges of climate change. Emphases on the efficient utilization of resources reduces the environmental degradation enhancing a cleaner and sustainable industrial

sector. Promoting sustainable practices among small and medium-sized enterprises (SMEs) will mean that they will increase their contribution to the global efforts in mitigating the impact of climate change.



Figure 6 SDG 13: Climate Action

Through the undertaking of small and medium enterprises with data-driven technologies, the project will be part of making a sustainable future in the manufacturing industry, creating efficiency in workplace operations, and improving environmental stewardship.



Figure 7 SDGs Mapping

1.8 Summary

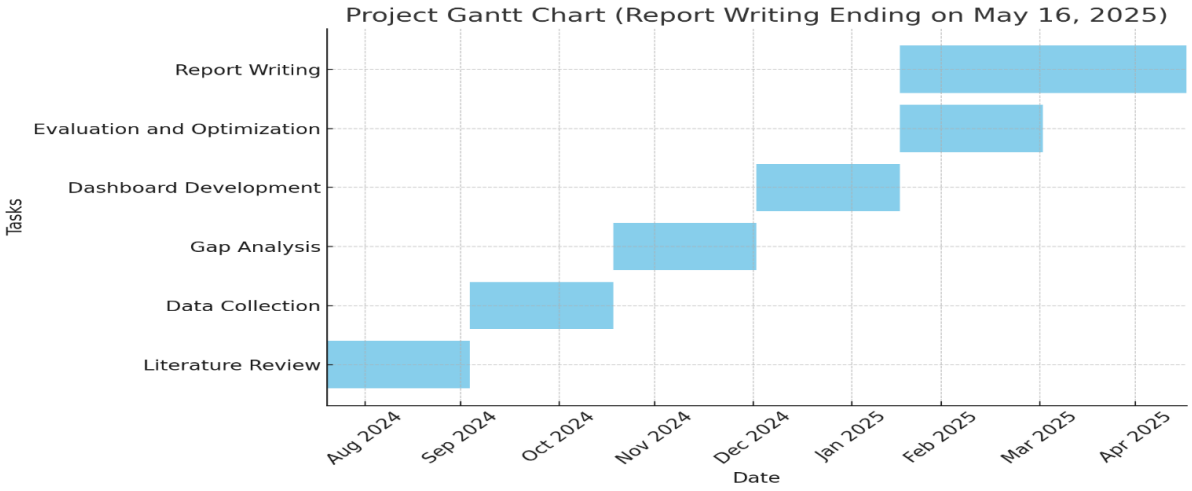


Figure 8 Project Gantt Chart

This chapter explains a framework written in Python that was designed to revolutionize quality management in manufacturing companies by carefully, in a coordinated way, integrating sophisticated methods, in particular Association Rule Mining, Causal Inference, and Counterfactual Analysis. The framework provides data-intensive proactive decision-making and sustainability, including basic issues, like the detection of faults, root cause analysis, and inefficient operations. The system is based on a well defined eight-phase methodology covering information gathering and pre-processing, classification of defects, discovery of patterns, establishment of root causes, and assessment of scenarios. Key applications include minimizing quality flaws, streamlining of resources, quality engineering of the supply-chain and sustaining resource-oriented programs. In line with the Sustainable Development Goals (SDGs), the project stimulates economic growth, innovation in the industry, and the responsible production behavior. As any implementation may include both costs of setting up and complexity of the procedures, the value of the improved quality control, savings, and data-driven decision-making are considered to outweigh the shortcomings. To conclude, the framework provides a solid foundation in building a strong, scalable and sustainable system of quality management that can help the industries meet their operational excellence and adjust to the changing industrial horizon.

Chapter 02: Related Work

This chapter follows the historical evolution of quality-control practice, the traditional tools that were used in the past and the methodologies that are nowadays fashionable (dependent on data). It outlines the main tools that became the pillars of the quality-control workflow and points out the ones used in the project under review, as well as explains the rationale behind such decisions as in **Figure 9**.



Figure 9 Flow chart Chapter 2

2.1 Introduction to Data-Driven Quality Management

Data driven quality management has changed the way things get done in small and medium sized businesses (SMEs), in recent years. These businesses have shifted gears of reactive quality management to proactive and predictive control by the implementation of modern technologies that look to Industry 4.0 ideology, thus ensuring competitive advantages and chances of sustainable growth. [1].

The impact of machine-learning algorithms such as regression, clustering, and neural networks has become a part of the necessity in the analysis of production data allowing identifying defects

and optimize the resources and making sensible decisions. The tools also speed up the process of timely detection and correction of quality anomalies by automating the monitoring process and, therefore, increase the efficiency of operations and the overall productivity[2].

Yet, the application of causal inference, promoted by Pearl in the Book of Why, is far beyond correlation because it can be used to form accurate cause-and effect relations in production systems. This kind of strategy is particularly beneficial to the small and medium-sized enterprises (SMEs), as it can help them grow the operations on the basis of data-driven decisions and address particular quality concerns. When SMEs use causal models, their defects prediction is changed to promoting a high level of the quality of the production.[3].

Counterfactual analysis is a supplement to decision-making, which organizes the search on other scenarios and allows predicting the implications of various approaches to quality management. The SMEs get the ability of being able to come up with the capacity of what they call sense what-if simulations, which in turn generate actionable insights capable of curbing the defects and also optimizing the allocation of resources through the Propensity Score Matching or PSM. [4].

Technologies of Industry 4.0, with the biggest role belonging to the Internet of Things (IoT) and cyber-physical systems, keep reshaping data-driven quality management. Through implementation of Internet of Things sensors, businesses are able to gain immediate data gathering thus, offering real-time details about the sites of production. The information can generate analytical models that are used by small and medium sized businesses (SMEs) to optimize the process of operations, manage the delays in throughput, and make autonomous management choices. In turn, such transparency is integrated in real-time dashboards and visualization instruments, promoting a steady process enhancement. [5, 6].

Digital Twin technology which is one of the key building blocks of Industry 4.0 will allow small- and medium-sized enterprises (SMEs) to create virtual representations of physical systems. These models allow simulation and optimization of manufacturing activities without having to shut down any daily operations hence causing less error and loss of time. Through such integration of advanced tools, SMEs can remain competitive and sustainable even in a very demanding market.[7].

Despite the considerable benefits of data-driven approaches, they have a significant range of counterarguments: high costs connected to utilize technological infrastructure and hire or retain a workforce with all the competencies required, as well as with the persistence of issues connected with data security. These constraints can easily reduce the large-scale adoption of the modern quality management techniques, mitigation of these constraints is thus essential.[8].

Table 1 Introduction to Data-Driven Quality Management

Reference	**Objective**	**Limitation**
[1]	Defect prediction and quality evaluation are done using machine learning methods (example with neural networks-based models).	Usage of such capabilities is limited to enterprises which are able to gather and process large volumes of data.
[2]	ARM is a methodology that was attributed to the establishment of the relationship between production factors.	There may be situations when the use of ARM architecture is insufficient in the following conditions: highly dynamic manufacturing facilities.
[3]	Causal inference is brought to explain cause and effect relations in production environments.	In possession of vast experience in statistical modeling and casual analysis.
[4]	It is possible to follow the decision-making by counterfactual analysis that entails building hypothetical scenarios where events progress in a different way.	The problem has a high computation complexity, especially the ones that facilitate parallel-search process, and it would need a lot of familiarity with methodologies like (PSM).

[5, 6]	Promoting ARM efficiencies with the evolutionary algorithms during manufactures.	Evolutionary algorithms are occasionally complicated by their very nature, an aspect that presents quite a challenge in actual implementation.
[7]	The Digital Twin technology can be deployed to optimize production processes.	Deploying an IoT solution requires high financial investment as well as the use of proper technical skills.
[8]	Predictive maintenance on data analytics would involve mining past failure to help predict future failure which would help perform preventive moves instead of repairs.	Computational biology requires a lot of computing power and dependence on skilled staff.

2.2 Tools Used in Traditional Quality Control (QC)

onventional Quality Control (QC) efforts have been the foundation of quality of product and process reliability in manufacturing companies throughout decades. These methods make it possible to conduct constant monitoring and gradual enhancement of quality results, basing on statistical inquiry, on-site visual assessment, and a formalized step-by-step instructions. Despite the fact that they have a necessary heritage, these methods are met by great obstacles in modern large-volume, rapidly changing production environments. Some of the major tools of classical quality control include Statistical Process Control (SPC), Pareto Analysis, Six Sigma, Fishbone Diagrams, Failure Mode and Effects Analysis (FMEA and Regression Analysis), Check Sheets.

2.2.1 Statistical Process Control (SPC)

Statistical Process Control (SPC) is one of the most used tools of the classic quality control is Statistical Process Control (SPC). It depends on control charts in tracking the variations of a process overtime and to identify emergent trends or anomalies that can indicate a pending quality problem. The method is based on the assumption that variation can exist as an inherent

part of any productive process, but the astute distinction between both natural variation commonly referred to as common cause together with unnatural variation commonly known as special cause is essential in retaining consistency. SPC charts provide a graphical representation of data where operators have an overview idea on when the process parameters become out of control. An example can be given of an automotive assembly line having a control chart which shows that there is a deviation in the bolt tension; a visual tip off like this would enable quick corrective action to be taken to prevent the manufacture and supply of faulty parts. Even though SPC is very good in the identification of trends, its manual interpretation and use of non-dynamic historic data limits its usefulness in high-volume or real-time applications[9].

2.2.2 Pareto Analysis

Pareto Analysis is a set of tools based on the 80/20 concept that is used as a conventional prioritization method in the scope of quality control. It determines those factors that contribute most to quality issue and this has allowed organizations to focus resources on the 20 percent of the issues that cause 80 percent of defects. In a pharmaceutical manufacturing facility, Pareto Analysis may verify in a pharmaceutical manufacturing facility that packaging error and contamination are the causes of most defects. The two areas raise a significant potential of increasing the overall quality. However, Pareto Analysis is simply descriptive as it is only focused on what is wrong, and it makes no explanation on the causes of the defects or even makes no predictions on future defects.[10].

2.2.3 Six Sigma (DMAIC)

Six Sigma is a data- based paradigm, which is aimed at controlling process variation and improving quality. As its structured process, DMAIC (Define, Measure, Analyze, Improve, Control), gives a systematic approach to quality issues. The discrete stages of DMAIC fulfil discrete functions:

- **Define:** Delineate the problem and articulate performance objectives.
- **Measure:** Record the data to get a foundation of the performance measure.
- **Analyze:** Determine the root causes of defects.
- **Improve:** Implement measures that can help solve the identified root causes.

• **Control:** Establish controls to sustain the improvements achieved.

Six Sigma has been of great use in critical areas like electronics where strict supervision of microchip manufacturing is a requirement. Still, the approach is dependent on the static historical data and statistical models which could not be aligned with the ever-growing needs of the dynamic, real-time production. [11].

2.2.4 Fishbone Diagram (Ishikawa)

A Fishbone Diagram, also known as Ishikawa Diagram is an organized brainstorming diagram which is used to determine likely root cause of defects. The diagram categorizes causes into six main groups, namely Man, Machine, Method, Material, Measurement and Environment. This systematic model helps teams look into several causes behind quality issues. [12]. In the sphere of textile production, uneven distribution of the dye may have as various origins as inefficient machine adjustment, errors in the competence of operators, or lack of quality fabrics. In spite of the fact that the Fishbone Diagram is suitable to generating the hypotheses, the limitation of efficacy is the dependence on subjective data and the absence of quantitative confirmation.

2.2. 5 Failure Mode and Effects Analysis (FMEA)

Failure Mode and Effects Analysis (FMEA) is a prevention approach that attempts to analyze the possibilities of occurrence of the failure mode, its impact on the quality of the product, as well as ranking the risks in terms of the level of severity, frequency, and detections. The fact that FMEA assigns Risk Priority Numbers (RPN) to each failure mode allows organisations to focus on the critical issues [13].

More often than not, failure modes and effects analysis (FMEA) in the field of aerospace manufacturing detects the risks associated with material fatigue in the component parts of related aircrafts. This is a critical step and essential toward reliability and safety mitigation of these risks. However, FMEA is rather labor-intensive and subjective practice; it requires a lot of time and effort to study the details of every failure mode.

2.2.6 Regression Analysis

Regression Analysis is a statistical method used to explore the relationship between the independent variables, e.g. temperature and pressure and the related dependent variable, e.g.

defect rate. This approach is commonly applied when the process parameters which have an impact on the product quality should be identified [14].

In a food processing plant, regression analysis can explain how variation in oven temperature caused the variation in homogeneity of a food product. Even though it is inevitable in explaining linear relationships, this method falls short when faced with the interdependent and non-linear variables that define many multi-faceted production systems.

2.2.7 Check Sheets

Check sheets refer to simple but extremely efficient tools of gathering data and documenting it using an orderly procedure. They become particularly helpful when some data has to be collected in the course of an inspection, an audit or even production operations [15].A check sheet can be used to list defects in various categories in real time as the production shift is going on. The tool is simple to use, enhances uniformity in data collection, and makes it possible to conduct an instant analysis. However, it records raw data and further tools of analysis are normally needed in order to derive meaningful information on the information gathered.

Table 2 Tools Used in Traditional Quality Control (QC)

Reference	**Objective**	**Limitations**
[9]	It is essential to keep on track changes of the process over time.	Mass-production systems that run in real-time have low utility to constrained operations.
[10]	It is necessary to rank the key aspects of any quality-related issue.	Makes the inquiry about the nature of defects without considering what causes them.
[11]	The focal goal of contemporary manufacturing is to lessen variation in the processes and enhance quality.	The statistical model in question makes use of only static, historical data.

[12]	Defect prevention programmes rely on particular identification of the possible root causes of defects.	The report under consideration is of subjective nature and lacks the quantitative validation.
[13]	With the help of RPN model, risks should be analyzed as well as prioritized according to their priority.	Time-intensive and subjective
[14]	Close analysis of links between process variables needs to be conducted.	The field of study tackles the non-linear interdependent variables.
[15]	Record and gather data in a carefully and systematically fashioned manner.	The current data set contains raw information that is not subjected to additional analysis.

2.3 Modern Tools in Quality Control

Transformation schemes driven by Industry 4.0 technologies have essentially reconfigured quality-control (QC) practices through integrations of real-time quality-control data analytics, automatization and advanced computing solutions.These tools do away with the limitations of the traditional QC providing greater accuracy, efficiency, and superior predictive capacity. The following discussion gives an elaborate explanation of the key modern QC tools. Processing of data in real time makes it possible to constantly monitor production processes to identify defects automatically. They allow analysing specific process variables on a discrete basis, which creates a possibility to detect anomaly in near real-time, thus, mitigating the task of expedited correction. In its turn, automation presents the technology to automate process tasks that had to be performed manually. Automatic functions can be implemented to perform functions like calibration, inspection and sampling and allow personnel to perform different functions of higher value. High-level computing practices are capable of providing vital analytical authority that cannot be matched by legacy systems. Pattern recognition and classifications can be

performed by means of machine learning, deep learning, and neural networks and are very complex and high-end techniques, thus allowing to improve decision accuracy in terms of QC. The integration of these technology in QC systems has accrued significant benefits in terms of accuracy, efficiency, and predictive power relative to the conventional processes.

2.3.1 Machine Learning (ML) and Artificial Intelligence (AI)

Intelligent automation has enabled machine learning algorithms to take a dominant position in the contemporary quality control setting. The systems are used to predict faults, optimize the working functions, and do some decision-making to automation. The manufacturers identify the patterns, observe the potential quality problems in advance and recommend the preventive corrective actions with the help of classification, regression and anomaly detection. ML therefore increases the accuracy of defect identification and inspiring process improvement. However, the performance of such models continues to be solely reliant on access to large sufficiently annotated datasets, and this limitation is shared by many industries. [16].

2.3.2 Real-Time Monitoring and Visualization Dashboards

The Internet of Things (IoT) systematically accounts the real-time data concerning equipment performance as well as the production conditions along with the quality of finished goods; by incorporating the sensors and devices to the production line. With such sensors, it is possible to conduct a constant watch on the manufacturing process and this provides prior information on a failure that could occur. The resulting IoT-driven dashboards and alert systems can thus respond and maneuver around possible quality slip-up promptly in order to minimize non-operating-time and maintain product uniformity. Such systemic transparency eventually increases the efficiency of the operations and enables the organization to maintain the bar of quality. However, the cost, which goes into the initial set-up and the fact that many organizations are struggling to integrate the IoT with the legacy systems can also be significant indignances [17].

2.3.3 Reinforcement Learning for Process Optimization

Reinforcement Learning (RL) is a subfield of the artificial intelligence and is specifically concerned with optimizing dynamic systems through sequential interaction with the environment of production. RL algorithms scan the changes and gradually optimize their tactics, sequentially reducing the level of defects, and improving performance. Such

functionality is especially advantageous in complex manufacturing settings. However, RL deployment is hindered by a high level of computation and the fact that it requires special technical skills, two aspects that may limit its use to a wider application[18].

2.3.4 Digital Twin Technology for Simulation and Prediction

Digital Twin Technology implies the production of virtual images of real systems capable of replicating and analyzing production processes. When this is used in conjunction with real-time data, possible failures can be avoided, what is to be expected can be predicted, and option settings can be made without putting a halt on everything that is currently ongoing. The ability enhances proactive decision-making, reduces idle time and ensures that the outputs produced are of high quality. However, construction and maintenance of the digital twins presuppose significant investments in the IoT infrastructure and modeling skills[19].

2.3.5 Blockchain for Quality Assurance and Traceability

Blockchain is a tamper proof, decentralised ledger system that furthers quality assurance through enhancing traceability. This technology promotes openness and responsibility, in that, by logging every level of manufacturing, counterfeiting should be discouraged, and, if there are defects in manufacturing, they should be easy to detect. Its usefulness is especially relevant in industries that have a high level of compliance, especially pharmaceuticals and food processing. However, the implementation of the blockchain requires a significant amount of computing power and equal amount of resource energy consumption[20].

Table 3 Modern Tools in Quality Control

Reference	**Objective**	**Limitations**
[16]	Automate defect detection and classification	The degree of reliability depends on the availability of high quality extensive lists of labeled data.
[17]	Monitoring dashboards provide real time information on activities.	Both high-cost setup and the legacy system integration are the challenges these face.

[18]	In a bid to maximize dynamic processes so as to increase general quality.	Technical hurdles and computational complexity are two of the major pillars entrenched in the science of computer science.
[19]	Simulate and predict process outcomes	Requires investment in IoT infrastructure
[20]	The amendments that should increase traceability and prevent the counterfeit.	High energy and computational costs

2.4 Tools We Used in the Project

The selected tools of this project are carefully chosen to overcome the weaknesses of traditional QC tools to use the potential of new, data-driven approaches. The tools chosen were meant to work in series hence facilitating effective, precise and solution-focused quality control program. The tools enabled quality management practices to be intensively improved within the small and medium-size enterprises (SMEs) by integrating the most recently developed analytics, the Machine learning and systematic workflows.

2.4.1 Data Collection and Preprocessing

Collection and preprocessing of data were the core pillars of the project. In this process, a wide body of data was collected including the parameters of production, the rate of defects, and operational factors. The quality of the obtained dataset was maintained by carefully upholding preprocessing steps. Systematic procedure was also used to fill-in the missing values, detect outliers, and resolve the inconsistencies. Meanwhile, the data were organized in such a way that they could be susceptible to further sophisticated analysis. The resultant dataset that would be both clean and ordered would be a product of the preprocessing operations thus likely to be easily integrated into succeeding machine learning models. It was not only that this preparation work made the subsequent analysis safe but also created the environment through which valuable insights can be developed and put into action.

2.4.2 Defect Classification by MLP

Classification of defects was a critical part of the current project; the objective was to classify merchandise based on the different levels of perceived quality. The Multi-Layer Perceptron (MLP) architecture was used in the project as it is one way of automating the process of placing each product in one of three bins that could be assigned to high volumes- defect-free, minor defects or major defects. This automated process was completely automatized thereby minimizing on the irregularities that arise with manual inspections thereby protecting the quality and homogeneity of the quality monitoring process. In addition, the models of classifications used exploited the available historical data in order to identify the key trends in the defect occurrence thus helping in the real-time classification of the modern items as they were moving along the production line. These early detection allowed early immediate countermeasures to be taken. [21].

2.4.3 Rework Reduction by MLP

The predictive quality analysis has injected proactive aspect to traditional quality-control scheme. Multilayer perceptron (MLP) models are applied to predict how many items will have to be reworked and estimate possible defects that will arise in the production process next. The models pass the judgment on the historical information and real-time information to isolate risk factors that can trigger quality problems. This vision allows optimization of the resources and shortens the downtime due to the fact that possible problems are addressed prior to their manifestation in the form of those acknowledgeable defects. Finally, the prospective plan will take the efficiency of the production up by a significant margin and curb wastage because quality issues do not become inflamed unnecessarily. [22].

2.4.4 Root Cause Analysis by APROTRI Algorithm

A deeper level of investigation of the production process occurred through the use of root cause analysis, where patterns and relationships between many process parameters along with the emergence of defects were identified. This probe was critical in a way that it contributed to identifying the real causes of quality issues. In order to study such complex interaction, APRIORI algorithm as an efficient analytical tool was employed to investigate the effect of machine settings, material properties, and environment parameters. Using this analysis, corrective measures were possible since it gave a clear understanding of the reasons behind the

faults. Root cause analysis therefore helped in ensuring that interventions were realised in a proactive way where the cause of defects was tackled and prolonged improvement on the quality of the product was realised.[23] .

2.4.5 Causal Analysis by Causal Inference (PSM)

Causal analysis is an extensive expansion of root cause analysis, which allows systematic explanation of deterministic cause and effects relationships between variables of the production and defects. It does not narrow investigation to correlations, as what determines quality outcomes are pinpointed as definite factors. In this regard, LINGAM and causal inference approach are the methods that can be used to test such associations. Having a concise understanding of causation, producers can prioritize interventions and provide their resources to the most high-impact issues and thus provide the maximum possible result of quality control as well as an efficacy of managing this process. Such specific and accuracy-based method yield considerable arithmetic gains in production stability and goods quality. [4].

2.4.6 Counterfactual Analysis by Causal Inference (PSM)

Together with causal analysis, counterfactual one was also used to explore alternative scenarios, with each scenario stating the potential outcome offered by the use of different approaches to quality management. Using a process-specific approach to modelling like PSM, instead of asking itself what-if questions, the study demonstrated how defect rates and overall production quality would change as production parameters, such as what machine settings, material inputs, or environmental parameters were used in production, changed. Such a future approach provided decision-makers with the knowledge needed to process improvements, resource allocations, and corrective actions. Counterfactual analysis prevented the scenario of suboptimal decision-making since it possible to try out different strategies before using them in practice and provide a reliable base on which the further quality improvements could rely.

Table 4 Tools We Used in the Project

Reference	Objective	Limitations
[21]	This project aims at automating the process of defect detection and classification.	the current study is based on high-quality labeled datasets.
[22]	The organization can have the ability of availing real-time predictive knowledge.	Expensive setup and integration challenges
[23]	Simplify operations in order to improve on quality.	Computational complexity and technical challenges
[4]	Establish cause-effect relationships	Requires extensive data and statistical expertise

2.5 Why We Used These Tools

In search of the stronger quality control, modern artificial intelligence methods, namely Multi-Layer Perceptron (MLP), APRIORI algorithm, and Propensity Score Matching (PSM) were carried. Such methods are more precise, cost effective and promote actionable intelligence that goes in line with the needs of small and medium sized businesses (SMEs).

2.5.1 Multi-Layer Perceptron (MLP) for Defect Classification

Defect classification is automated in MLP and scans trends on the production data hence eliminates the slowness and error susceptibility of manual classification. Its ongoing real-time inspection can classify the products as no defects, mild defects, and serious defects and guarantees both speed and quality uniformity inspection. [24].

2.5.2 Multi-Layer Perceptron (MLP) for Rework Reduction

The MLP forecasting predicts the rework requirement based on the analysis of the past performance and real-time performance to identify the possible risk factors. This proactive

approach minimizes waiting, wastages, and minimizes capability to direct resources to maximize overall output through efficiency. [22].

2.5.3 Root Cause Analysis with APROTRI Algorithm

The APRIORI algorithm offers a logical approach toward defining the fundamental causes of product defects noticing the links between machine settings, material qualities, and process factors. This type of analysis will make it easier to develop accurate corrective actions, therefore, helping to eliminate the flaws where they get nurtured and causing sustainable changes in product quality. [23].

2.5.4 Causal Discovery Using LiNGAM (Linear Non-Gaussian Acyclic Model)

LINGAM specifies the causality between production variables and defects and thus resolves the drawback of using correlation-based methods. It helps to implement the next specific intervention, and it allows the achievement of significant quality improvements as it highlights such significant determinants as the calibration of machines and the quality of material. [4].

2.5.5 Counterfactual Analysis for Scenario Evaluation

In addition to the traditional method of causal analysis, the remaining study incorporated the Counterfactual Analysis, which was used to build hypothetical decision-making processes, predicting their outcomes. Allowing the estimation of such a situation as changing the material feeding in or the regulation of the machines, the meeting allowed the project to simulate future changes of the defects frequency and the entire quality of the products. With Prospective Safety Methodology to validate such simulations, counterfactual analysis allowed such an accurate picture of which generation was the most efficient in the process of improving production and thus offered the needed perspective to proactively act to improve quality-control through advances in such strategies. The methodology therefore defended the decisions relating to quality-control with the help of predictive models, which reduced the threats of unexpected results and enhanced the capacity of the project to make decisions in the long-run. [4].

2.6 Summary

The chapter examines how data-driven approaches may supplement quality control in small and medium-sized enterprise (SME) to complement the deficiencies of the past quality management approaches. The methods include machine-learning implementations, root-cause analysis

algorithms, and causal-inference processes which work together in enhancing efficiency, accuracy, making informed decisions. A stringent practice in data gathering as well as preprocessing ensures that manufacturing documents are in impeccable condition, reliability, as well as being susceptible to complex analytics. Through the Multi-Layer Perceptron (MLP), the problem of classifying defects as well as predicting the rework surfaces; the problem of proactive quality control finally becomes possible. Examining past data sets and tracking current events, multilayer perceptron models develop an understanding of existing trends and risk factors, which excludes the necessity of manual action and improves the overall standards of efficiency. APRIORI algorithm is the key to root-cause investigation because it outlines the relationships between parameters of production and defect incidences and allows carrying out highly specific corrective actions. Similarly, methods of causal-inference like Propensity Score Matching (PSM) go even further than correlation and establish a verifiable cause-effect relationship, thus making sure corrective measures are taken to resolve the most important quality issue. The systematic quality management paradigm, which is provided through the highly integrated array of these instruments, offers a look into the future. Dropped rates of defects, alleviated rework, and finer decision-making are jointly a contribution to greater reliability of production, fewer wastes and long-term competitiveness. However, there are obstacles (such as computational needs, specific know how and the necessary investments in infrastructures) to be overcome to achieve widespread usage. All in all, various contemporary and data-driven responses that are found in this project have the power to transform quality management at the face of SMEs.

Chapter 3: Methodology

In this chapter, the project methodology included in this book defines the stages that are ordered sequentially by the project, as well as support data-driven decision-making with the data collection and preparation, data analysis, and data evaluation. It starts with the discovery, selection, and cleaning of the information, then, arrives at the implementation of machine learning algorithms, and ends with root cause and causal analysis. It also contains a counterfactual analysis and finishes off with the presentation of the results of the project as shown **Figure 10**.

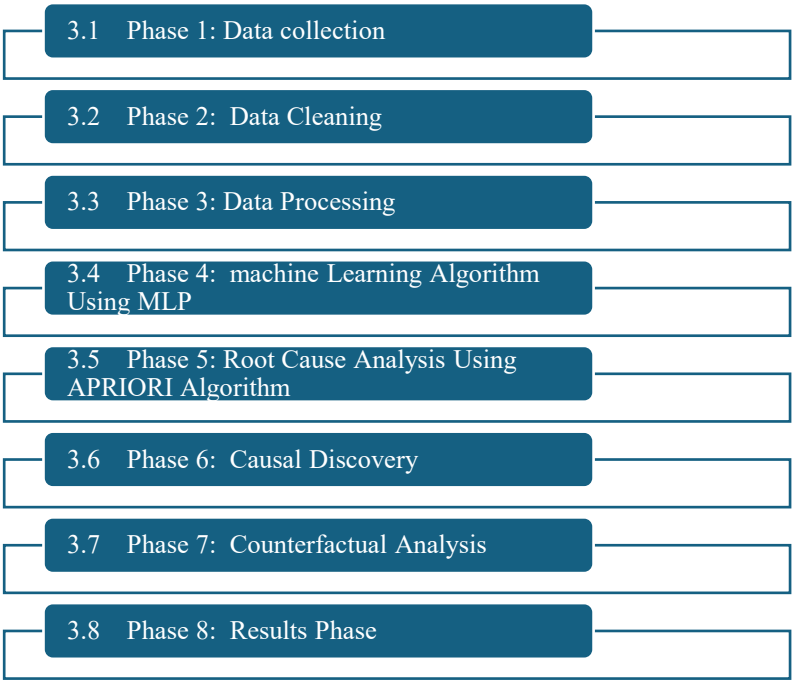


Figure 10 Flow chart Chapter 3

The chapter provides precise the use of Big Data Analysis in the Quality Control (QC) of the Small and Medium-sized Enterprises (SMEs).

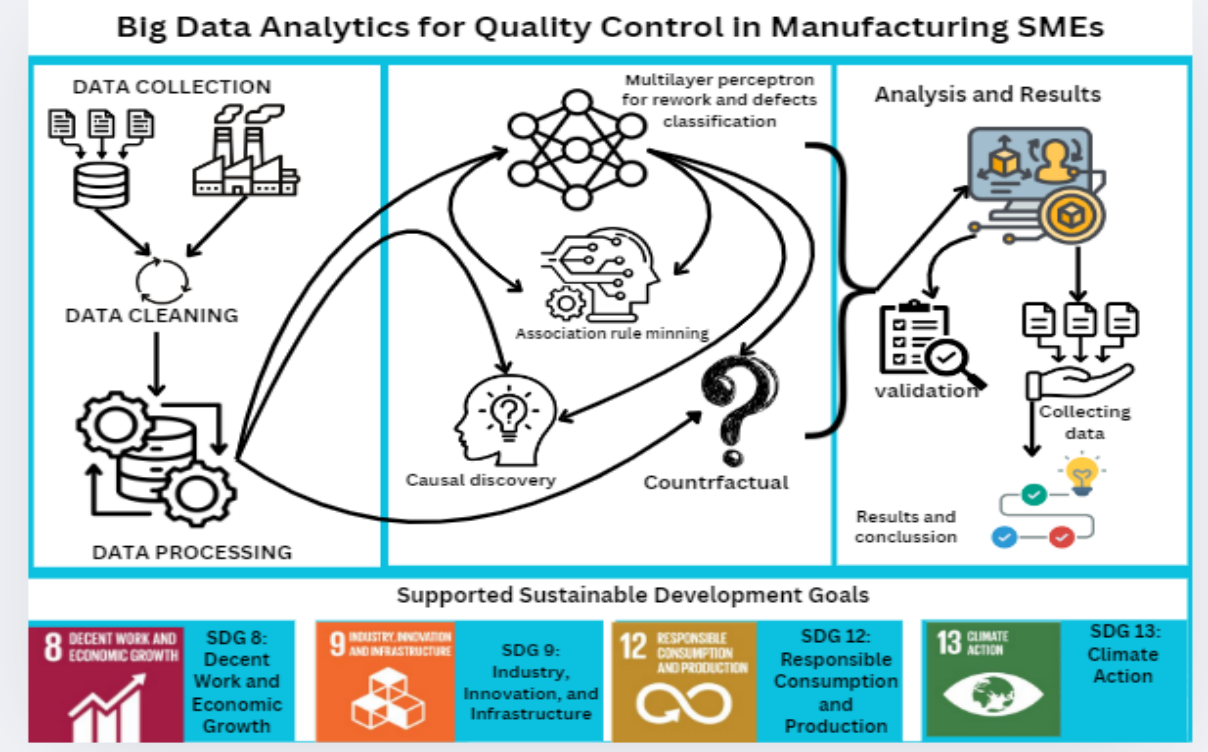


Figure 11 Methodology Diagram

The procedure is structured in such a way that it has distinct steps that ensure the data collection, processing, analysis, and interpretation steps follow a disciplined flow..

- Phase 1: Data Collection
- Phase 2: Data Cleaning
- Phase 3: Data Preprocessing
- Phase 4: Machine Learning Analysis Using MLP
- Phase 5: Root Cause Analysis Using APROTRI Algorithm
- Phase 6: Causal Discovery
- Phase 7: Counterfactual Analysis
- Phase 8: Results Phase

3.1 Phase 1: Data Collection

The Data Collection step proves to be a critical engagement related to the implementation of Big Data Analysis in Quality Control (QC) of organizations based in small and medium-sized enterprises (SMEs), because it ensures the availability of high-quality data provided by multiple

sources to serve as foundational material to achieve analysis and to provide the suggestions to be implemented. The information is obtained through the deployment of IoT sensors in key production points hence real time data on machine output, defects and parameters of the production is captured. Complimentary manual logging gives us extra qualitative data which in automatic mode are not captured. The Historical enterprise databases add continuing directions and lines of business to be individualized by assessment, whereas external environmental information, such as temperature and humidity, brings into account things encountered outside the manufacture setting which impact on the quality of the product. The surveys of the employees reveal procedural inconsistencies as well as operational hurdles and thus, the automation analytics are complemented by the frontline views. At the same time, customer feedback systems collect quality replies, defect detection reports, and provide external assessment of QC success. The reliability of data collection and reliability is based on a systematic way of doing it. IoT devices have deliberate placements, manual records are computerized, and instantaneous measurements are recorded at a certain frequency that perfectly coincides with production cycles. Mechanics are put in place to identify and resolve data gaps, inconsistency and errors as well as periodical audits and cross checks to ensure data integrity is enhanced. This phase will build the necessary platform towards making meaningful analysis taking into consideration accuracy and completeness; thus, the methodology will attain the QC objectives of SMEs.

3.2 Phase 2: Data Cleaning

The most important thing to do in the Phase 2 of Data Quality Control procedure is to ensure that the dataset that was gathered is correctly identified, is internally consistent and reliable enough to enable intensive analysis. To reach this goal, a variety of measures aimed at improving the quality of datasets and eliminating potential error origins are required. Based on the nature and extent of lost data, interpolations, imputations, or simple ways of excluding them are done. The checks are also conducted to ensure consistency as possible variations in the units of the variables, format, or the naming patterns will be identified and addressed to maintain the unanimity in the dataset. Distorting outliers, statistical processes (such as z-scores or interquartile ranges) can also regulate them; or where the situation demands it, domain-specific approaches may be used. Besides, unnecessary or irrelevant observations such as those

composed of duplicated entries or those beyond the scope of the analysis are filtered in a defined and orderly manner to reduce the noise and retain only the relevant information. The integrity of data is the main priority during this phase as well. Policies are put in place to ensure that each datum item plays a unique role in the analysis and eliminates redundancy and coverage of the dataset. Validation procedures ensure that the collected data has remained intact through its cleaning and sensitivity tests performed to gauge how cleaning decisions affect general results. The transparency is maintained by recording each process of cleaning so that there is an in-depth view of changes and improvement made to the dataset. It is commonplace that automated cleaning scripts or validation checks are in use in which efficiency and scalability are guaranteed. In this step, transparency in communication with stakeholders is promoted to explain the reasoning behind the decision regarding data cleaning and what implications lead to such decisions. Specific focus is given on the specific modifications that are made on the data, and this enables the stakeholders to clarify issues or doubts that may emerge. The result and outcome of this stage are an efficient, trustful set of data which provides a strong foundation upon which preprocessing and analysis would be done. The reliance on the quality assurance and transparency help to reinforce the integrity and success of the whole Quality Control framework, strengthen it with the help of the Data Cleaning stage.

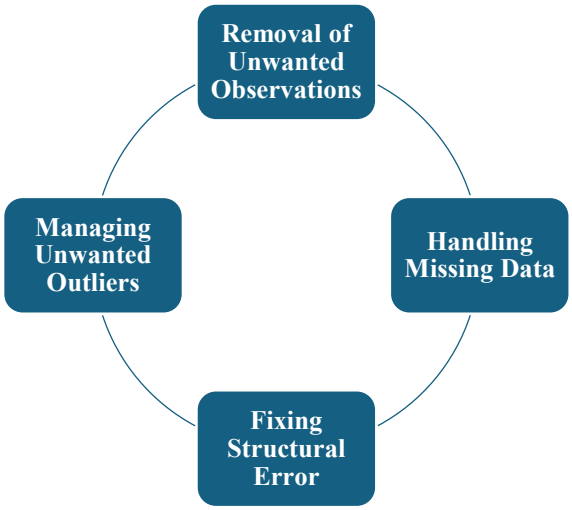


Figure 12 Phase 2: Data Cleaning Phase

3.3 Phase 3: Data Preprocessing

The Data Preprocessing stage that takes place once the dataset is cleaned up represents the precondition to the advanced analysis, modeling. At this phase, raw data are transformed into organized, normalized and combined forms that can support machine learning models, and statistical functions. Generalization methods can be used to normalize numerical gauges onto a similar scale thus making difference data sources compatible to one another. This is done to eliminate biases that are presented by heterogenous units or magnitudes. At the same time, feature engineering is the key element, which creates an additional set of variables that are supposed to improve the predictive capacity of the dataset. Examples are calculation of defect frequencies, proportion of machine use and measures of environmental impact. Those based on various sources, which include IoT monitors, physical logbooks, and third-party environmental data stores, are joined into one coherent database. This kind of integration ensures consistency of data through the removal of redundancies and makes it easy to analyze. Where relevant, transformations of time-series are performed, where raw time stamps are converted into interesting temporal features both across trends as well as seasonal patterns. Automated preprocessing pipelines are developed to handle great amounts of data quite effectively and reduce the need for manual interventions. These systems have validation checks which help in protecting data integrity in every step. Specification of the preprocessing steps in detail ensures that the studies are reproducible and traceable, and these two elements are necessary to ensure scientific rigor. The process of standardization, enrichment and consolidation of the dataset at the Data Preprocessing stage renders the latter ready to be used in downstream processes such as machine learning, statistical modelling and informed decision-making. As a result, it creates a vital chain linking raw data and actionable knowledge, steeps the ground of comprehensive Quality Control analysis.

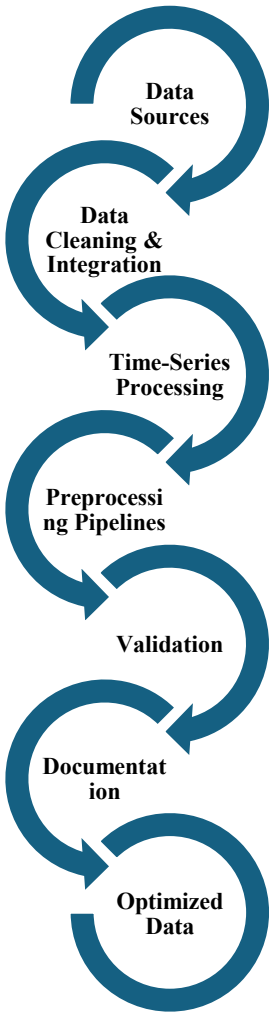


Figure 13 Phase 3: Data Preprocessing

3.4 Phase 4: Machine Learning Analysis Using MLP

This is because the Machine Learning Analysis phase makes use of the Multilayer Perceptron (MLP), which is an advanced neural learning method, to enhance Quality Control (QC) through defect classification and a predictive analysis. It transforms pre-processed data into actionable information through the application of the ability of MLP of identifying complex pattern and correlation present in the data. One of the main tasks performed by MLP is Defect Classification that automates the process of separating products into specific quality classes. This eliminates the element of subjectivity and inconsistency that manual scrutinizes carry out, therefore making quality measurements uniform and accurate. A machine-learning model is trained with

historical data with the aim that the model will be able to identify patterns of defects. After the training of the model, the model is tested upon unknown, independent data sets to determine model accuracy as well as model generalization. There is also predictive modeling that is used to predict possible defects and reworking requirements. MLP would determine spots with high risk of occurrence by analyzing past records and their occurrence in real-time and would send preventative measures to those areas. The ability to make accurate predictions yields three outstanding advantages, which include reducing wastages, maximizing resource utilization and increasing efficiency of production. Initiating the use of MLP involves data partition between the training and testing data, and sequential optimization of the model to deliver optimum results. The effectiveness of the model is determined by using such metrics as accuracy, precision, recall, and F1 scores. These metrics provide a wholesome explanation of the strengths and weaknesses of the model. This stage thus brings out the impactful possibility of machine learning on Quality Control, which allows SMEs to move towards proactive control of quality. The phase achieves major gains in reducing defects, the efficiency of operations and eventually a higher quality of overall production through the use of MLP. MLP diagram is shown in **Figure 14**.

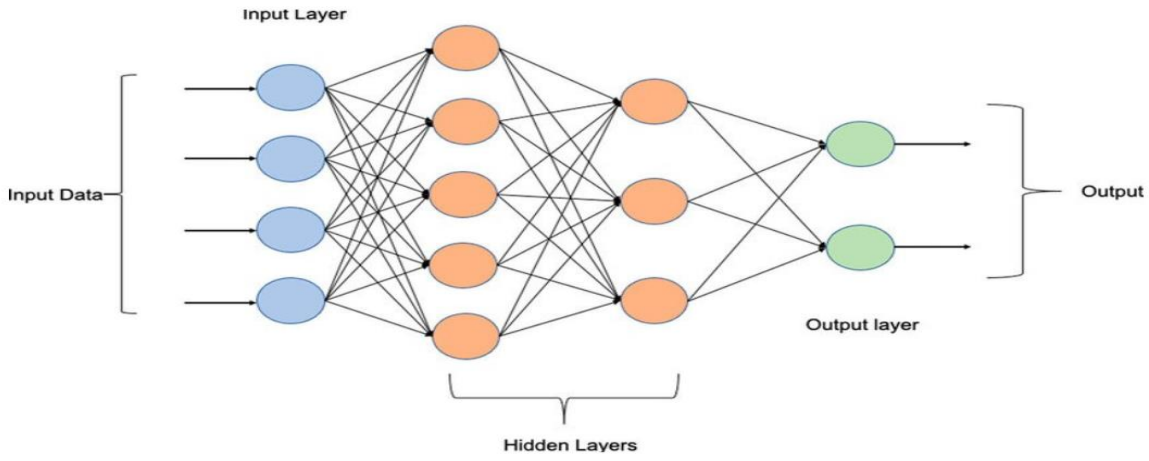


Figure 14 Multilayer Perceptron Diagram

3.5 Phase 5: Root Cause Analysis Using APROTRI Algorithm

The Root Cause Analysis stage uses an APRIORI Algorithm, which is an efficient analysis tool, to reveal the causative factors behind the creation of quality issues in production. Such a stage is essential as it will allow recognizing and correcting root causes instead of treating symptoms

and only facilitating long-term quality changes. APRIORI uses advanced pattern-identification methodologies and statistic research to identify connections amidst the production variables, such as machine adjustments, material expressions, and climate situations. By means of the systematic study of such interdependencies, the algorithm brings to light such factors that have the significant impact on the departure rates. The resulting knowledge allows focusing corrective actions on most important variables. The process starts with integration of well-formatted and cleaned data on to the APRIORI framework. APRIORI will then perform a series of iterative analyses that will enable isolation of variables with a view of quantifying the effect of the variables on the occurrence of defects. Validity of the output is achieved through comparison with past results and consultation with subject-matter experts and hence reliability and relevance of results. Where there is root causes identified, there are interventions to be created to counter the effects. To reduce defects, it might be possible to use adjustments to the calibration of the machines used, material chosen or timing of the process. Besides tackling short-term issues, APRIORI provides information on long-term optimization of the manufacturing process. By specifically focusing on the actual causes of defects, this step significantly improves production quality as well as levels of efficiency. The implementation of the APROTRI algorithm makes the intervention data-driven, accurate, and effective and, therefore, creates a more stable and reliable Quality Control framework.

Consider a collection T of transactions, formally defined as $\{T_1, T_2, \ldots, T_n\}$ [25], where each transaction T_i is a subset of a set I of distinct items, i.e. $\{i_1, i_2, \ldots, i_m\}$. Given two disjoint subsets X, Y I, an association rule can be written as X Y, where: X is called the antecedent and Y the consequent.

Measurement of the utility of association rules is typically carried out with the help of three main measures:

- **Support:** Refers to the percent of transactions in which a set of items occurs.

$$\text{Support } (X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

The support for the rule **X → Y** is calculated as:

$$\text{Support}(X \rightarrow Y) = \text{Support}(X \cup Y)$$

- **Confidence:** It is an indication of how reliable a rule is as it indicates the probability of event Y given the instance of event X:

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

- **Lift**: The current analysis seeks to determine the level of association between two variables, X and Y, in one of which having a lift value greater than 1 is seen as an indication of the existence of a positive relationship.

$$\text{Lift}(\text{X} \rightarrow \text{Y}) = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(X)}$$

In research article, the Apriori algorithm, is one of the most standard approaches to association rule mining, was used. Apriori determines frequent item sets and, therefore, creates association rules by defining user-set stream of support and confidence. Apriori then uses the downward closure Apriori, an itemset is frequent when all the itemset of shorter length (n-1) are frequent as shown **Figure 15**. This property saves a lot of computing time. Despite its efficiency, Apriori algorithm are facing two challenges but due to its strong performance in handling large datasets and identifying useful association rules, Apriori algorithm still has wide applications in data mining literature.

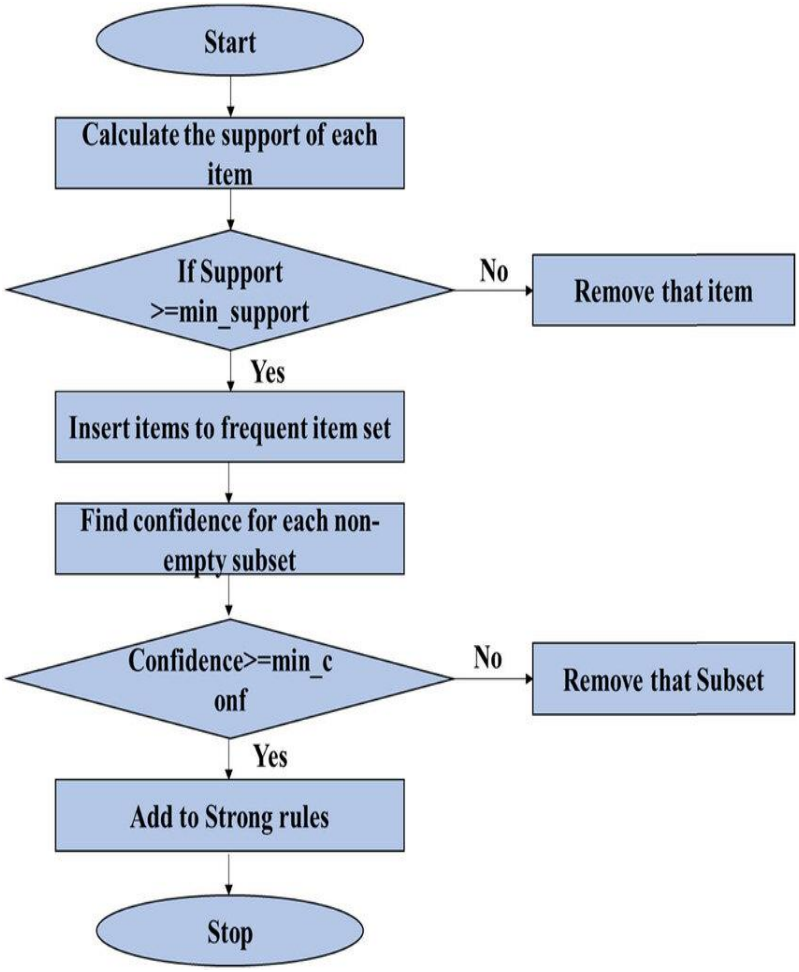


Figure 15 Apriori Algorithm Flowchart

3.6 Phase 6: Causal Discovery

LiNGAM technique (Linear Non-Gaussian Acyclic Model) is a method of estimating and building a causal model in data. Linear Non-Gaussian Acyclic Model is a method of statistics that supports nonlinear independent errors and is the basis of the introduction to socio-economic modeling applications. It works using Directed Acyclic Graphs (DAGs), i.e. constructs, the members of which are a set of functions with no direct cycles or feedback. Moreover, LiNGAM lacks any constraints and in the past years has proved to be a new strategy to define causality without the need of experimentation. This technical property also makes the approach particularly beneficial to economics, neuroscience and artificial intelligence domains where non-stationary perturbations are common..

Linear Non-Gaussian Acyclic Model:

$$x_i = \sum_{k(j)<k(i)} b_{ij} x_j + e_i \quad \text{or} \quad \boldsymbol{x} = B\boldsymbol{x} + \boldsymbol{e}$$

- $k(i)(i = 1, \dots, p)$: causal (topological) order of x_i
- Errores variables e_i independent and non-Gaussian
- Coefficients and causal orders identifiable
- Causal graph identifiable **Figure 16**

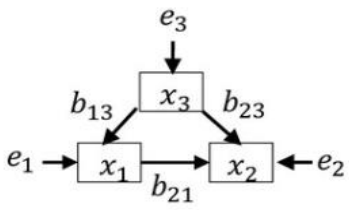


Figure 16 *Causal graph identifiable*

3.7 Phase 7: Counterfactual Analysis

The counterfactual analysis used in this project will be to evaluate other decision-making possibilities and their resulting outcome on manufacturing-quality results. In this phase some of the what-if conditions are simulated to consider the variation in the production variables like the machine settings, the use of either material, the environment in which the final product can be produced, that in turn can affect the overall quality of the final product. In the observational case setting, the counterfactual analysis methods have been well known to be effective when estimating the counterfactual effects. ([25]; [26]). Propensity Score Matching (PSM) is a statistical technique that is explicitly premeditated to address selection bias by advantageously assigning witnessed covariates in the treated and non-treated populations in a way that is fair. The process matches units with similar values of the propensity score, thus forming similar groups in estimation of the cause and approximating a randomized controlled trial.

Now suppose D is given by a binary treatment indicator defining D=1 to refer to treated units; D=0 to refer to untreated units. The response variable will be Y and X = {X_1, X_2, …, X_p} denotes a collection of covariates. The propensity score, e(X) is interpreted as the probability of being under treatment given the covariates.

$$e(X) = P(D = 1 \mid X)$$

The propensity score is used to just match treated and untreated units especially based on likelihood of receiving the treatment.

The use of matched groups is involved because of the intent of comparison; with the help of matched groups, the effect of a treatment on its outcome can be approximated.

To provide an in-depth evaluation of the effectiveness of PSM, two key measures are regularly used:

- **Average Treatment Effect on the Treated (ATT):** the distinction in results that are viewed amid treatment units and matching with common set of controls.

$$\text{ATT} = E[\,Y(1) - Y(0) \mid D = 1\,]$$

- **Average Treatment Effect (ATE):** A stringent evaluation of the effect on the outcomes of treated and untreated units is normally subjected to the estimations of a specification of the mean difference between the units.

$$\text{ATE} = E[Y(1) - Y(0)]$$

PSM involves the following steps:

1. **Estimating the Propensity Score:** to find Propensity Score we use a logistic regression or machine learning model for each unit based on covariates.
2. **Matching:** After calculating Propensity Score of a particular unit, it is matched using nearest neighbor matching.
3. **Assessing Balance:** After performing propensity-score matching, a more elaborate scrutiny of covariate balance is done to establish more thoroughly the similarity of the produced cohorts. This verification is habitually carried out using standardized mean differences (SMD).
4. **Treatment Effect Estimation:** Causal effects are usually evaluated by calculating the average or median difference between the group of people that was being treated, and compared relative to a cohort of people that were untreated.

Effectively performed propensity score matching (PSM) allows appropriately mitigating the effects of the observed difference covariates which makes it extremely applicable in supporting analysis of observational data. Still, PSM does not cure confounding factors that are not observed and thus is susceptible to attack because of the selection of matching approach and

the caliper-width. Despite these limitations, PSM has remained in case of use in social sciences, medicine as well as economics in an attempt to derive credible causal implications using non-randomized information ([27];[28]).

3.8 Phase 8: Results Phase

In the Results Phase, the products of the previous phases are summarized, which allows comprehending the scope of the project, as well as offer measurable products that can be used to improve QC activity in SME. The main task of this stage is to present the findings in a succinct and strictly interpretable form so that the information can be utilized to make informed choices, and continuous improvement of the process may occur. Among the most significant results acquired with multilayer perceptron analysis, it is possible to outline the areas of high accuracy in the defects classes, be in an accuracy and recall.With the help of the predictive models created during the previous phases, the estimation of future defect rates as well as rework amount is made accordingly, which allows proactive interventions. APRIORI Algorithm-based Root Cause Analysis targets the key variables that determine the defective outputs and gives prioritized recommendations on what to do to remedy the situation. On parallel lines, the Causal Analysis using LINGAM identifies causative factors and determines the degree of their impact on the quality of production so that resources are allocated optimally. The results are shared using a complete set of visualizations such as defect trend graphs, heatmaps and decision matrices, which encourage understanding by intuition. Stakeholders obtain the useable steps of how to optimize processes, allocate resources and long-term strategies of how to improve quality. Together, the Results Phase incorporates all the steps of the methodology into a convenient model of quality improvement. With the help of this data, SMEs would have the opportunity to pursue a proactive, data-driven QC paradigm, significantly decreasing the level of defects and enhancing the total efficiency of production.

3.9 Summary

The given chapter describes a well-designed approach to incorporating Big Data Analysis into the Quality Control of small and medium-sized businesses. It is easy to see a logical sequence of steps throughout the analytical cycle including the collection of data, its cleaning and preprocessing, followed by analysis, all of which is necessary to obtain actionable insights.

Multi-Layer Perceptron, APRIORI Algorithm, and Propensity Score Matching tools will also be used by defects with precision, causal inference, and root-cause analysis in this process and will deliver indestructible solutions to the common QC issues. The approach enables SMEs to graduate to proactive quality management by helping them use sophisticated analytics and machine learning to achieve efficiency in production, the reliability of their products, and data-based decision-making. Finally, this methodological approach highlights the sustainability of the companies and their competitiveness in the fast-paced industrial environment in the long-term perspective.

Chapter 4 Case Study

The current chapter can be viewed as a case study of the Five-Stage Analytics Framework that was developed as a quality-control tool to be applied in the manufacturing SMEs. The case is based on a factual information of a medium sized denim garment manufacturing plant. Despite the advanced equipment that the company uses and strict procedure of the inspection, 4-6 % of persistent defect rates resulted in high costs of reworking. To address this problem, this company tapped high-order machine-learning and data-mining algorithms. The framework also includes five major components, in the first part, Multi-Layer Perceptron (MLP) models are used to both classify the severity of defects and predict the percentage of defects that will require a rework; second, Association Rule Mining (ARM) using the Apriori algorithm is applied to reveal the latent associations between defects and their causes; third, Causal Discovery using Linear Non-Gaussian Acyclic Model (LiNGAM) will identify the real causal relations and not just associations; fourth, Counterfactual Analysis with Propensity Score Match The methodological decisions are driven with the criteria including accuracy, scalability, interpretability, and robustness. MLP classifier classifies defects as Minor, Moderate or Major based on the quantitative type of defect characteristics hence giving a detailed outlook on the levels of product-quality. Another MLP model is based on rework time and cost on whether a defective item should be reworked or rejected. ARM allows to discover the hidden associations among defects and causes hence focusing on specific maintenance and training of operators. LiNGAM sheds light to causal connections underlying these relationships and the smarts based within these relationships. Lastly, PSM carries out counterfactual assessment of planned corrective measures, which is informative on preventive plans. Harmonized implementation of these methods will enable the SMEs to shift their focus in the area of quality-control out of the reactive quality-control operations that they have conventionally been implementing towards a proactive data-based quality-control solutions that would cut their defect rates, the rates of rework, and the generally inefficiencies in their productions among others.

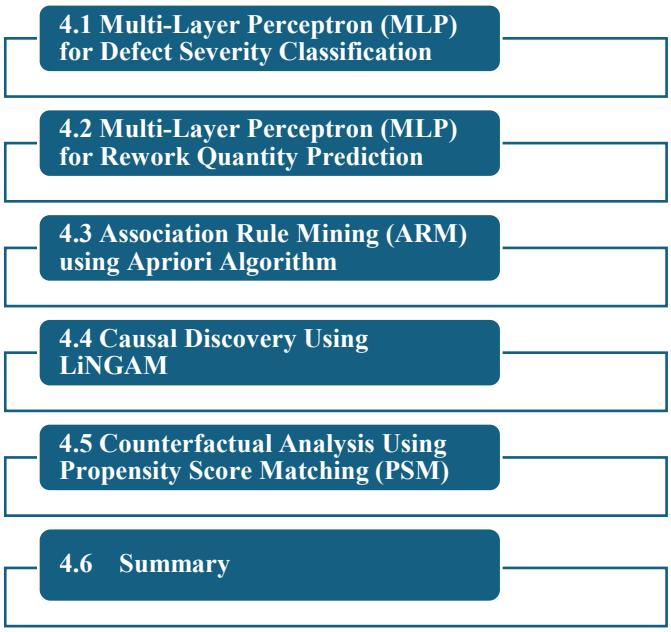


Figure 17 Flow chart Chapter 4

4.1 Multi-Layer Perceptron (MLP) for Defect Severity Classification

4.1.1 Overview of Multi-Layer Perceptron

The multi-layer perception (MLP) is a strong kind of feed-forward artificial neural networks and is commonly used in supervised learning activities including classification and regression. It has three main parts which are input layers (which process the original data), a hidden layer (in which the computations and upstream extraction of features are carried out) and an output layer which provides the final predictions. The neurons in each layer are connected to all the neurons in the next layer through weighted connections and activation functions providing nonlinearity, which allow the network to simulate complex, nonlinear relationships. In the case study, the MLP will be used to classify defects in denim garments into three levels of severity, that is, Minor, Moderate and Major. It is with the help of the learning capacity in MLP that we are able to achieve the goal of replacing the conventional manual classification process with an automated one which would be fast, objective, and scalable and thereby ensure improved precision and consistency in the quality-control process.

4.1.2 Data Collection and Preprocessing

The study is based on information gathered at a denim clothing manufacturing business which

specialises in the manufacture of jeans to be sold to the international and the local market. The paper focuses on eight production defects that are commonly found in Manufacturing: Broken Stitches, Seam Puckering, Thread Breakage, Buttonhole Defects, Misaligned Pockets, Color Bleeding, Zipper Defects and fabric damage. The number of defects present in each encounter was rated between 1-10 showing the severity of a given defect. In order to make the data ready to be used in machine learning, the following data preprocessing steps have been performed. To begin with, all the features were standardized using the StandardScaler method, hence, making sure that all the features have a mean of zero and standard deviation of one, and hence, preventing the biasness that would have accrued due to the different scaling. Secondly, the classes of severity (Minor, Moderate, Major) were converted into numbers (label encoding) to prepare the target variable to multi-class modeling with classification. The resulting clean and well-organized data can then be used as a good base to train the model.

4.1.3 Input Features and Target Labels

In the case of Multilayer Perceptron (MLP) model, the data comprised the standardized values of eight canonical types of defects-which are discrete dimensions of the garment quality-as the multidimensional input vector to all samples. The target label was the severity of the defect in the categories, Minor Defect (class 0), Moderate Defect (class 1), and Major Defect (class 2). The multi-class set up allowed the network to recognize subtle trends at various intensities of defects and resolutions thereof, thus enabling correct judgement of the quality of garment products.

4.1.4 Model Architecture Design

The architecture of the MLP model determines whether it works and performs or not as well as its ability to generalize. During the current research, the network and its architecture were intentionally created to achieve the optimal trade-off between the complexity and accurateness of training. The first layer was an input layer of 8 neurons, one representing each input data of the features. They were followed by three hidden layers, that is sixty-four, thirty-two and sixteen neurons respectively. Both layers used the Rectified Linear Unit (ReLU) as the activation function as it speeds up the training process and it reduces the vanishing gradient problem. The output layer was the final layer with three neurons reflecting the three classes of the severities of the defect, and their output was converted into absolute probability distribution on the three

possible classes with Softmax activation function. The architecture of this MLP allows it to store complex interactions of features easily maintaining an added level of computational feasibility.

4.1.5 Model Training Configuration

The model being proposed was built with carefully selected hyperparameters with the purpose of making the learning process faster and avoiding overfitting. Backpropagation was carried out in the supervision of the Adam optimizer, an algorithm that got a great deal of popularity with its adaptive learning-rate qualities and its ability to be reliable in various tasks. The loss function used was categorical cross-entropy that is a good choice when the desired output is a probability distribution and the number of classes is an arbitrary number. Mini-batch gradient descent used the batch size of 8, which allows achieving a balance between fast training and steady convergence. The procedure of 50 epochs was performed that provided a balance between having exposure to the data and being susceptible to overtraining. Twenty percent of the training instances were reserved as validation and generalization performance could be constantly measured during training. The choice of hyperparameters did not consist of any novelty: the batch size, number of epochs, and the relevant optimizer, which we chose, were based on initial experimental findings and well-documented best practices in training neural networks.

4.1.6 Performance Metrics and Evaluation

The outcome of the multi-layer perceptron (MLP) model was measured with statistical measures of accuracy and visually coupled loss and accuracy curves. They both rose gradually during training and ended up at a very high accuracy: more than 99.13 percent, with the minimum deviation between the two curves as well. This overlap implies that there was no overfitting. In addition, the accuracy of the final test was outstanding with the value of 99.03%, evidencing the strong generalization ability of the model. The low and constant loss of value during the training and validation stages strengthened the stability of the model. A confusion matrix also indicated that MLP had few misclassifications on the Minor, Moderate and Major classes. Its value and suitability in industrial defect classification applications can be revealed in the high scores of precisions, recall and F1, which were achieved because of models' outputs.

4.1.7 Practical Applications in Manufacturing

A number of tangible benefits are gained by incorporating a multilayered perceptron (MLP) model into the operations of an industry. Importantly it will automatically classify the severity of defects; also reducing the number of human inspectors and their subjectiveness. Its abilities to do real-time classification enable responsive action to be taken, therefore, reducing response times and minimizing wastage. Moreover, robots do not get tired, which improves the efficiency of production, and an outstanding consistency of quality. The classifier output information can be useful in higher-level decision-supporting systems which can be used to determine activities like predictive maintenance, resource optimization and strategic planning actions. Finally, the ability to deploy objective parameters in the classification of defects positions an organization to achieve better levels of customer satisfaction by being in a position to deliver products consistently with high rates of quality as shown in **Figure 18.**

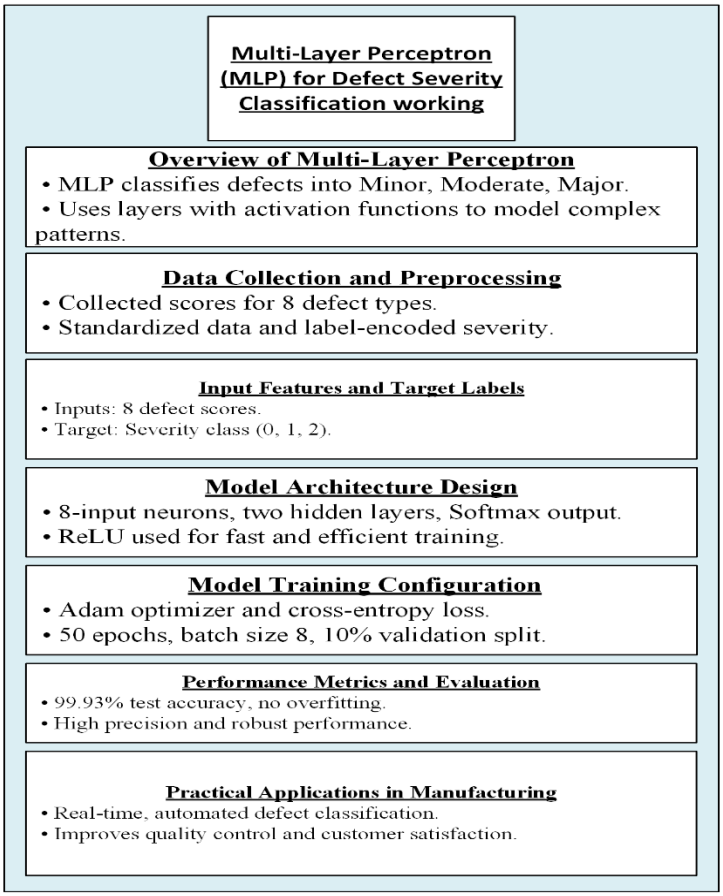


Figure 18 MLP Model Flowchart

Table 5 Advantages and Disadvantages of MLP

Advantages	**Disadvantages**
Obtains a very high precision of multi-class classification.	In this method, that is, sufficient training requires a large amount of labeled data available.
Capable of determining and describing complicated, non-linear trends of the sets of data.	Hyperparameter sensitivity: especially learning rate, and amount of neurons.
Adam optimizer and the rectifier linear unit (ReLU) activation feature are known to produce quick convergence rates in the deep learning exercises.	Works in a black-box manner, and therefore making interpretability to be a problem.
Easy to implement in real-time production scenario.	The amount of control required computing resources can be extremely huge with exceptionally large datasets.

4.2 Multi-Layer Perceptron (MLP) for Rework Quantity Prediction

4.2.1 Objective and Importance of Rework Prediction

In any production system, the landmark decision is whether to rework a defective product or reject the whole piece. Prediction in Rework is a vital tool of decision making that has a direct impact on the efficiency of the operation, cost containment, and project delivery. By way of an example in the case of denim jeans manufacturing, some defects may be handled within the reasonable cost and time frames whereas there are some others that make reworking not economical. As a result, proactive quality management is achieved through an automated system that has the capabilities of predicting rework of an item or its rejection. The current discussion attempts to design a binary classification model based on multilayer perceptron (MLP) so that it can recommend the right action of rework or rejection to be taken based on the defect attributes. With this important decision made automatically, the factory will be able to

reduce a lot of waste, avoid wasting money on disappointing labor and facilitate production planning.

4.2.2 Data Preparation and Feature Engineering

To compile the dataset devoted to the rework prediction the same denim production line was used, focusing on the clothes with some defects that may be reworked. Three major variables were chosen: the level of defect severity-Minor, Moderate or Major- estimated cost of rework, and estimated time of rework. These attributes were selected due to the fact that they have a direct influence on the rework-versus-reject decision at a financial and an operational perspective. In a bid to make the data ready to be modelled each feature was standardized using the StandardScaler method to ensure that each feature would contribute equally in the learning process. The dependent variable was categorical (binary): all those items, which were found suitable to be reworked, were assigned class, 1 (Rework), and those items, which were found to be too expensive or very time-consuming to repair were being assigned class 0 (Reject). Two threshold rules could be selected based on domain expertise: in case the defect was classified into Minor or Moderate category and the sum of costs and time spent on reworking the product presupposed a certain amount of tolerance, the product was reworked; otherwise, it was rejected. The given pipeline resulted in the clean, balanced data thus making the pipeline suitable to supervised binary classification.

4.2.3 Binary Classification Strategy

- The current research treated the rework prediction as a classifications problem of two classes. In this regard, the model was to be used in classifying every defective product as either belonging to either of the two classes:
- **Class 1 (Rework)**: Defected product can be fixed with efficiency in cost and time perspective and as such, it can be reworked.
- **Class 0 (Reject)**: A damaged product would require repair costs that disproportionately high or use an extravagant amount of time and thus make not fit to be used any more and suitable to be disposed of.

The limitations of output to two discrete classes make the decision-making process on the production floor much simpler and ensure the direct actionability of the output of the model. The clear benefits of binary classification in this scenario are that the derived predictions can

be understood by those at the shop-floor who would be supervising the quality control systems, and that the derived model can be easily incorporated into manufacturing execution systems (MES) to enable on-line quality control.

4.2.4 Model Architecture and Hyperparameter Settings

The rework MLP model was small but versatile. The building was designed in the following way:

• **Input Layer:** There are 3 neurons in the input layer, an input feature neuron to each of the three input features that are the defect severity, rework cost and rework time.

• **First Concealed Layer:** 16 neurons whose activation is ReLU. The resulting layer introduces only many parameters that are needed to capture complex feature interactions to an adequate depth level.

• **The Second Hidden Layer:** 8 neurons, ReLU activation of them, too. The second layer smooths the patterns obtained and by the first layer.

• **Output Layer:** 1 output layer, containing 1 neuron and Sigmoid activation function, gives value between 0 and 1 which is the representation of the probability that a particular item should undergo rework.

The model was assembled using the adaptive learning rate and stability in training Adam optimizer. It was based on binary cross-entropy loss where the loss is suited to the binary classification problem because the prediction contains a probability. To enable mini-batch updates and to converge in a much quicker manner, the batch size was set to 8. The training epochs were adjusted to 50, which was enough time to train but keep eye on the validation loss so as to prevent overfitting. The train-test split also stayed at 80:20 and this ensured that both training and testing of the model was done on a balanced dataset.

4.2.5 Model Evaluation

Various measures used to evaluate the performance of the model consist of accuracy, precision, recall and confusion matrix. The training and validation accuracy started with high values which means that model generalization was done well on initiation. The training and validation accuracy also increased steadily throughout the epoch and eventually stabilized at nearly 99.13% thus showing that the model has learnt well without overfitting. The test accuracy after the final testing was also a good result of 99.08 % accuracy and so the model was almost able

to perfectly foresee on the rework or rejection of the defective item. The confusion matrix was analyzed to show high true positive value (reworks predicted correctly) and a high true negative value (the rejects were predicted correctly). More importantly, no false rejections were identified, i.e. no item which was to be reworked was in fact rejected, and this is most desirable at operational level. Only a minute number of false positives was met, i.e. there were a little number of items forecasted to require rework that would have been better to reject, which is also an acceptable value. Training and validation loss curves were near the bottom and consistent, which further proves the point that there was no overfitting. Where manufacturing is concerned, this predictive precision is useful in making a cost-effective decision, minimizing human error and improving production lines.

4.2.6 Industrial Implications of Rework Prediction

The use of the automated rework prediction model, based on MLP, has major ramifications to a process of production. To start with, it eliminated dramatically the use of subjective human judgment that can differ depending on inspectors and time of shift. On the contrary, decisions can be based on objective data working continuously throughout the production process. Secondly, factories are able to reduce wastage by anticipating unequivocally those garments that may be effectively reworked and achieve significant savings in materials as well as labour. This is also good against the objectives of sustainability since there are reduced cases of unnecessary discarding of garment. Thirdly, it is possible to plan production using an automated prediction model. By prior knowledge of the ratio of the rework to the reject items, the supervisors are able to properly mobilise the rework resources hence avoiding bottlenecks and delays. Further, it has ensured that only garments that pass the quality standards are shipped with low quality products being filtered steadily. Strategically, these systems incorporate resilience in the manufacturing process, enabling these processes to respond more appropriately to changes in quality and consequently continue to keep the standards of output very high. Long term benefits can be easily achieved because the companies adopting such predictive models will be able to gain competitive advantage because of the increased operational efficiency, decreased costs of reworks and also the increased reputatio due to product quality as **Figure 19.**

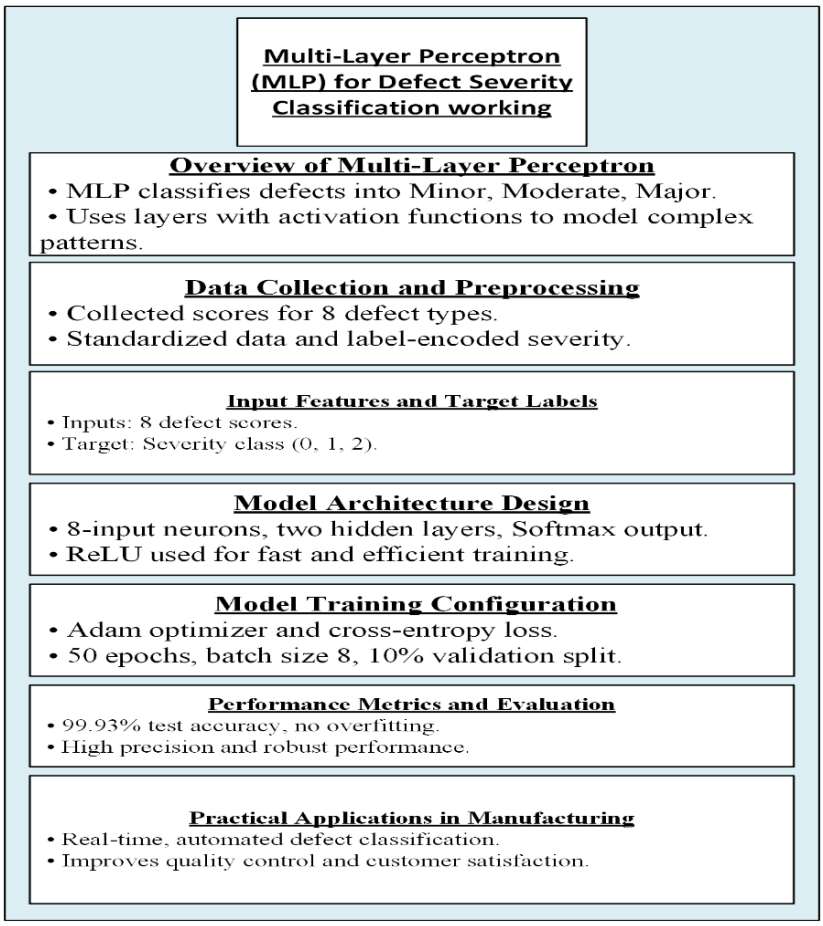


Figure 19 MLP Rework Prediction Flowchart

Table 6 Advantages and Disadvantages of MLP in Rework

Advantages	**Disadvantages**
Extremely high levels of accuracy and reliability of a binary decision-making.	Needs steady update of data in order to maintain its accuracy.
Rejects the human error in the decision of rework.	Might require retraining in the case of substantial change in profiles of defects.
Rapid real time forecasts in order to take instant shop-floor decisions.	They may need resources to deploy the model as well.

Minimizes wastage and also streamlines reworking.	Sensitive to initial hyperparameter selection
Increases the performance of the entire manufacturing operations.	Giving meaning to decision boundaries is conversely more handicapped.

4.3 Association Rule Mining (ARM) using Apriori Algorithm

4.3.1 Overview of Association Rule Mining

ARM or association rule mining is a strong data mining process, which is commonly used in order to identify interesting relationships, concealed patterns, as well as correlations amongst the variables of big data marks. What ARM does, especially in the case of SMEs involved in manufacturing, is to associate defects with the potential reasons and provides feasible information with which to enhance the quality control process. ARM applies two fundamental processes, the first is that the identification of frequent itemset (sets of items that often appear together) and secondly, the generation of association rules that indicate the influence of the appearance of one set of items on the appearance of another. The main robustness of ARM is hidden in its capacity to unveil unexpected patterns that cannot be easily perceived with naked eye. In manufacturing businesses, knowing which part of the operation is most likely to cause a certain defect may allow efficiencies in preventative maintenance programs, training requirements and production lines. Apriori algorithm, a traditional algorithm of ARM, works on the principle that any subset of frequent itemset should also be frequent. It minimizes the number of itemset that have to be analyzed, hence maximizing the computing time. One of the most credible and fundamental algorithms of data mining is Apriori mainly on structured data of a transactional nature.

4.3.2 Data Transformation into Transactional Format

The data has to be converted into a transactional format before it can be analyzed using Apriori algorithm. Raw datasets in manufacturing SMEs typically have continuous parameters of the operational units and defect records that are recorded numerically or as a text. With ARM however, the data should be in binary form with each row a transaction and each column an item (no presence or absence of a specific cause or defect). In case study discussed, the information regarding production process of a jeans manufacturing company was converted

into binary transactional information. Every check on the garments was considered as a deal. Such variables as the following, "high thread tension," "improper stitch type," or even "broken stitches" were given as ones when it was detected and as zeros when not. This binary formation helps the Apriori algorithm to read and process the data easily, and hence, it is able to scan combinations of causes and defects in an organized manner. This change is essential since ARM flourishes on the identification of co-occurrence patterns the treatment aims at finding mixture that is relatively common in the transactions, it is directed at finding probable correlations between manufacturing errors and their underlying causes.

4.3.3 Frequent Itemset Generation

The most obvious computational problem in the Apriori scheme is the production of frequent itemsets. Frequent itemset is a lists of combinations consisting of one or more items (defects or causes) in the dataset that co-occur with a frequency greater than a preset minimum support measure. The minimum value of the support was given as 0.1 in the research which meant that a given itemset will be considered to be frequent only when it occurs in at least 10 percent of the overall transactions. Apriori algorithm successively identifies all itemsets (and, in turn, all combinations of those sets) that satisfy this criterion by starting with itemsets made up of single items (1-itemsets) and then (recursively) of subsets of items (2-itemsets, 3-itemsets, etc.). As an example, the frequent itemset formed by "high thread tension" and broken stitches, is common in various inspections of garments. The frequency generation of itemsets is a reduction of the space available and there is no chance that the system takes into account the trivial or non-statistically significant relationships in formulating rules. This is an extremely fast step since it makes use of the downward closure property that says; the bigger the itemset, the more frequent the subset of this itemset must be.

4.3.4 Rule Formation (Support, Confidence, Lift)

After identification of frequent itemsets, the second step will be to build association rules. Each rule is cast in the form of, e.g., IF X, THEN Y, that is, a given presence of causes or defects raises the probability of some others. The effectiveness and applicability of every rule is determined in terms of three principal measures namely support, confidence, and lift. Support shows the frequency in which the rule can be encountered in the complete data, and gives an idea on how significant it is. Confidence is a measure of how likely it is that Y will happen

given that X has happened which is a measure of reliability of the rule. Lift measures how strong the relationship between X and Y may be and the values greater than one imply that the relationship is positive. Such rules in this research were generated by minimum confidence of 0.5 and lift threshold larger than 1. One of the interesting rules found was the following: Broken Stitches, Buttonhole Defects High Thread Tension, which had the maximum lift of 1.0181 and showed good correlation between the defect in stitching and the incorrect tension values of the thread in the manufacturing process.

4.3.5 Evaluation of Association Rules

The derived rules were considered not only based on support, confidence and lift but also in terms of interpretability in the practical context and in terms of operational applicability. Rules which are found out should be relevant logically in the manufacturing process. As an example, it was revealed that the issues of inappropriate marking when fixing the pockets to the pocket attachment were highly connected with misaligned pockets as well as bleeding of the color. Indeed, those associations confirm recognized practical concerns on the aspect of garment production, poor marking in which may indeed directly affect the accuracy and appearance of stitching. Lift charts and support-confidence scatter plots were utilized to gauge and determine the best 10 rules. Rules that were too low in lift, or had very little logical sense, rules that did not measure up to the numerical values, were also rejected. Such combination of statistical expertise and domain expertise guarantees that the quality improvement programs feed only on actionable insights. The analysis took into consideration also the strength and breadth of the rules, rules that implicate multiple causes of multiple defects being found specially important, since they point to systematic problems of the production and not to individual ones.

4.3.6 Practical Insights from ARM

Third, the implementation of ARM results provided profound knowledge on the quality issues affecting the manufacturing unit: Systematic reason like excessive tension of the threads and mishandling of fabrics were identified to elicit a series of defects simultaneously. The wrong marking patterns of the sewing related operational habit had led to cascading defects, including pocket placements as well as even shade variations. Dependencies between machinery aspects such as bent or dull needles and flaws such as seam puckering and breakage of thread were well defined and hence the schedule of maintenance could be readjusted based on it. Active

countermeasures may be elaborated as an example: the operators may be retrained in fabric handling methods, and the equipment may be inspected with the emphasis on thread tension adjustment in order to reduce the most expensive defects. Using ARM results, managers may find it convenient to allocate resources to those improvements that have the most impact. Rather than running at random with troubleshooting, data driven efforts can now be made, which save time, cut production expenses, and it increases the overall product quality. The understanding that some defects have a tendency to be a part of a common group or that they have a common cause allows SMEs to shape evidence-based not only in results but also actual-based preventive maintenance plans, specific training programs, and ways of variable processes optimization as shown in **Figure 20.**

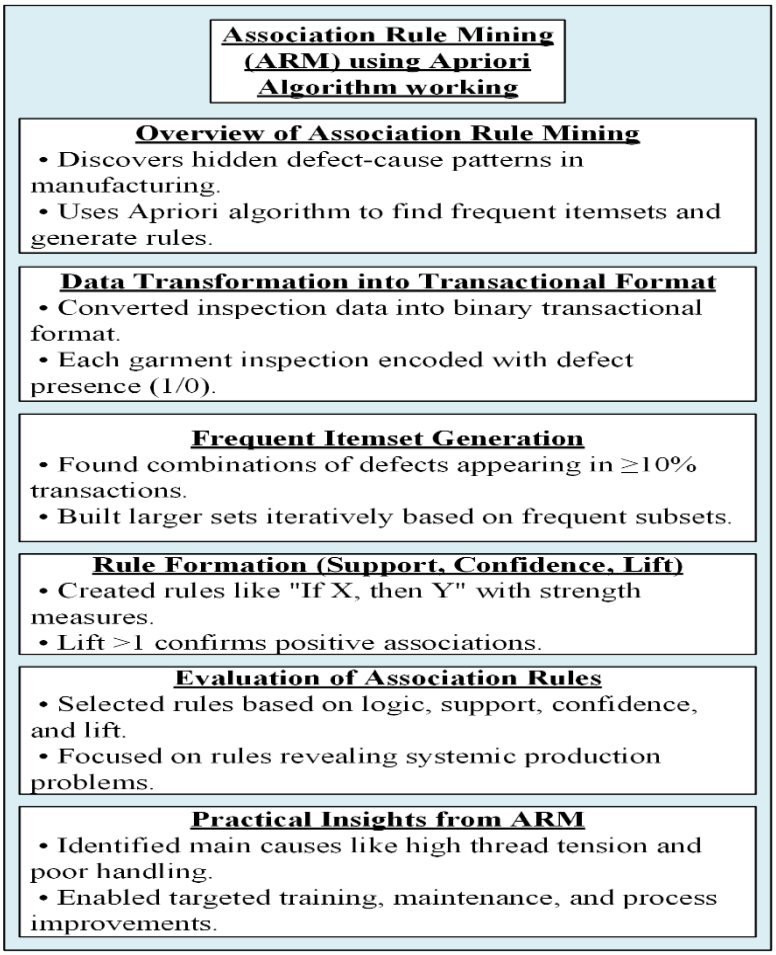


Figure 20 ARM Process Flowchart

Table 7 Advantages and Disadvantages of ARM

Advantages	**Disadvantages**
Reveals latent patterns which are not seeable by manual inspection.	Produces excess rules that lack the appropriate support/confidences levels.
Easy and straight forward results.	Very large scale computationally related.
Assists in finding the systemic causes to various defects.	Findings are not caused by links but co-relations.
One is the flexibility of thresholds to tune in on most important patterns.	Domain knowledge is required to enforce to check whether the rules are meaningful or not.
Increases the quality assurance and maintenance preventive measures.	Over simplification of complex process may be done through binary format transformation.

4.4 Causal Discovery Using LiNGAM

4.4.1 Introduction to Causal Inference

Causal inference refers to the identification and interpretation of the causeand-effect relationships or correlations between the variables and not mere correlations. Although correlation shows that two variables are associated with each other, it does not establish why, the relationship exists or how. In an industrial environment, especially among the manufacturing SME, it is of great essence to know the difference between entanglements and actual causations in order to apply relevant interventions. The traditional statistical tools are usually insufficient as they rely mostly on the idea of correlations. Here causal inference enters the scene, providing the possibility to answer what-if questions that are very important when it comes to quality control. Being able to find out the real causes of defects in the products manufactured helps the managers to not only be aware of the defects but also the causes and how to avert them in future. A scientific approach to making decisions causal inference methods

introduce clear, practical knowledge about the causes of manufacturing problems, which in the long term will contribute to proactive quality management approaches.

4.4.2 Fundamentals of LiNGAM Model

The Linear Non-Gaussian Acyclic Model (LiNGAM), is a causal discovery algorithm specialised to overcome the pitfalls of the traditional correlation-based methods of analysis. LiNGAM was developed by Shimizu et al. in 2006, and it is based on the assumptions that data generation mechanism might be expressed via a directed acyclic graph (DAG) and that the error terms incorporated should be non-Gaussian and statistically independent. Such assumptions make the LiNGAM unique among other approaches, because it can be used to uniquely determine causal structures without requiring structural interventions. The model expresses the associations among the variables in a linear manner, it however, withstands signs of non-normal data distributions; therefore, the model is especially adapted to real-life industrial data that seldom advance perfect statistical trends. LiNGAM is effective as it does not only tell the relationships between variables but also the direction of their influence so it can answer the question is it the high reason or the stitches that are broken. This is crucial to the quality control in manufacturing because failure to identify the cause and effect relationship could result to the ineffective corrective measures.

4.4.3 Data Preparation and Normalization

Prior to the use of LiNGAM, data preparation and normalization are necessary preparation processes that will make causal discovery precise and valid. As shown in the jeans manufacturing case study, operational parameters and defect attributes were initially made standard in order to eliminate scale biases. Standardization is standardizing all the input variables, so that any variation in the measuring units or the range of the values does not bias the analysis of the cause effects. This is important since LiNGAM works on linear models but any variance in the scales of the variables may provide false information to the algorithm concerning magnitude and direction of the causal ties. Also, the redundant, irrelevant or missing data were eliminated in the dataset and only the most relevant factors related to the operations and defects were used in the analysis. Such a pre-processing of the data serves two purposes, improving the efficiency of computations, and also the interpretability of the resulting causal graph, making it more likely to capture the true causal relationships in the manufacturing

process and not a spurious artifact of low data quality.

4.4.4 Directed Acyclic Graph (DAG) Construction

The basic structure that LiNGAM outputs is that of a Directed Acyclic Graph (DAG) so as to illustrate causal relationship between variables. (e.g., in a graphical model) nodes are variables (e.g., of the form, high thread tension; etc.), and directed edges or arrows express the direction of causation between one variable and another. The graph should not have any circular levels of causation, and running along the arrows in any one direction will not bring one back to beginning, in other words there is no looping. As in the manufacturing study, after the dataset was normalized, the LiNGAM algorithm produced a visual representation of how varied causes of defects were produced by the various factors of operation. As an example, inappropriate handling of fabric was indicated as the cause of not only misaligned pockets but also puckering of seams. Awareness of such relationships, by making them visible, allows the manufacturing managers to get the most important cause-effect paths in a short time, prioritize their interventions, as well as address the issues that impact positively within the group or grouping of the most influential factors that affect the quality. The DAG does not only summarize visibly but also acts as a strategic guide to decision-making that aims at reducing the defects.

4.4.5 Causal Relationship Estimation

Once the DAG has been constructed then the estimation of the strength of the causal relationship between each pair of variables should be done. LiNGAM provides coefficients that demonstrate the extent to which the variable affects the other variable. The larger the coefficient the greater the causal impact. An example is that given that the causal coefficient of high thread tension on thread breakage is 0.34, then the changes in thread tension directly affect the chances of thread breakage dramatically. The casual relationship estimation undertaken in the jeans manufacturing case study identified such factors as bent or dull needles, wrong stitch type and the wrong size of the needles as some of the leading causal factors in causing defects. Additionally, such causal chains that influencing stitching operations in improper fabric handling were found via intermediary routes of causing multiple defects were also observed by the model. The direct and indirect cause and effect pathways provide a more insightful understanding of the relationship between various areas of operation in the business thereby enabling SMEs to know where to take corrective actions at reduced costs.

4.4.6 Root Cause Identification and Effect Size Interpretation

The last and maybe the most important result of using the LiNGAM-based analysis is the identification of the root causes and the interpretation of their effect measurements. Root causes are operational or material conditions, which being eliminated or otherwise reduced would remove or greatly lessen a variety of defects further down the line. Some of the causes of the defect formation which were identified in the case study were high thread tension (0.34 coefficient), wrong fabric handling (0.34), bent or dull needles (0.31), and wrong stitch types (0.27). Interpretation of effect size is also instrumental regarding actions prioritization: the bigger the effect size the higher will be the increase of the product quality after dealing with its cause. As one example, several forms of defect can be reduced at the same time by concentrating on tension settings of the thread and maintenance of the needles. On the one hand, factors having weaker effect sizes represent an important issue but can be approached in the future or

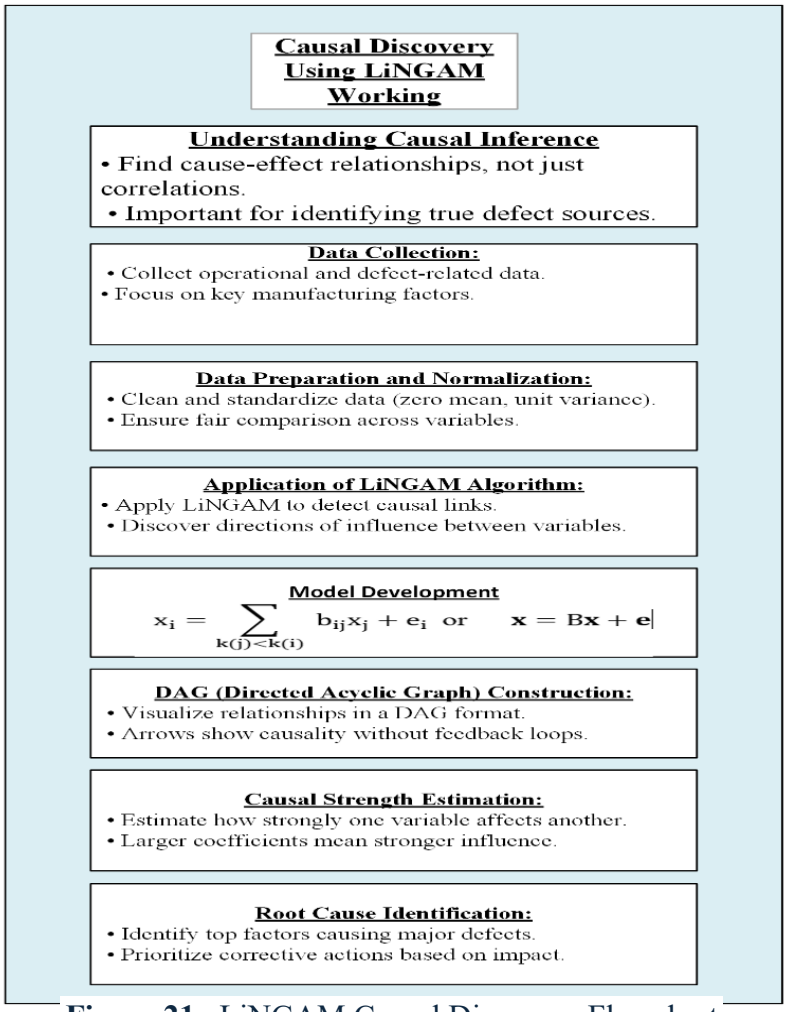


Figure 21 LiNGAM Causal Discovery Flowchart

whatsoever. Interpreting effect sizes in the DAG, manufacturers obtain a prioritized, quantified list of intervention points that can provide a transition of quality control approaches, previously described as vague, to extremely specific and data-provided interventions.

Table 8 Advantages and Disadvantages of LiNGAM

Advantages	**Disadvantages**
Determines actual cause-and-effect situations and not mere correlations making it possible to make better corrective measures.	Hypothesis with rigid prerequisites like linearity and non Gaussian errors, which are not always true with real life data.
Identifies the direction of relationships, as well as the question on which relationship depends on other.	Be sensitive to outliers and noise in the data which can give an incorrect causal structure when this is not correctly cleaned.
Does not require randomized experiments, as such it is appropriate with observational industrial data.	Can be difficult to learn causal models with very high amounts of causal complexity such as feedback loops, since it makes the assumption that causality is acyclic.
Gives measurable effect sizes that can assist to prioritize interventions depending on how great they could be.	Very large dataset is computation intensive, consuming and demanding a lot of processing resources and optimization.
Provides visual results (DAGs) that can easily be interpreted by a strategic decision-maker.	Even the understanding of weak causal connections might need topical knowledge not to be misused.

4.5 Counterfactual Analysis Using Propensity Score Matching (PSM)

4.5.1 Concept of Counterfactual Reasoning

Counterfactual reasoning is an effective tool of analysis, which helps to understand what could have happened with modified circumstances. In quality control, both identifying the causes of the defects as well as the consequences that would arise in case certain measures were changed are essential in manufacturing. Counterfactual analysis is a method that allows scholars and

practitioners to develop hypothetical scenarios and ask them the question of whether the defect rate would have been reduced had there been proper adjustment of thread tension. This provides the ability which helps to estimate the effects of the changes without the interference with the existing production that saves resources, time and budgets. This scope that transcends both correlation and causality provides a predictive foresight to the counterfactual inquiry hence making it more likely to provide adequate prevention measures. The approach is particularly useful to small and medium scale businesses, which are limited by the lack of capacity to conduct large scale experimentation or trial and error stagings. A particularly feasible approach, the Propensity Score Matching (PSM), tends to match the treated and untreated observations in observational data sets in such a way that to imitate an experimental study of randomized control, the reliability and validity of the subsequent results are improved.

4.5.2 Propensity Score Estimation

Propensity-score estimation is the central precondition of utilization of propensity-score matching (PSM) approach of counterfactual analysis. A propensity score is the likelihood that a unit, which could be a garment, an individual production batch, or a similar unit receives a treatment conditional on a number of observed covariates (operation setting, material estado, and environment, among others). In production environments, an exposure, or in this case a treatment, may be characterized as exposure to a production state; an example would be: high thread tension or an inappropriate manner of making stitches. Propensity scores are routinely generated using logistic regression, and an indicator variable of treatment (treated versus untreated) is defined as a dependent variable, and all relevant covariates form the independent ones. These scores must be estimated with great exactness as this ensures that the comparison drawn between the units being treated and the ones not treated is just and objective. On the other hand, miscalculation would interfere with the matching process hence rendering the analytical validity. After calculating the propensity scores each unit will be given a score between 0 and 1 depending on its probability of taking the treatment as a consequence of its individual attributes.

4.5.3 Matching Treated and Control Groups

After the estimation of the propensity scores, it is then the time to match the treated and control units according to the propensity scores. The main goal of matching is to create the balanced

data where the proportion of covariates is close between treated and untreated units. A variety of methods might be used, such nearest neighbor matching, caliper matching or radius matching. The nearest neighbor matching was also used in the current research and every treated unit was matched with the untreated unit that had the closest propensity score. This method helps make a fair comparison of the two groups, pretending them to be done in a randomized controlled experiment as the bias of selection will be reduced. Accordingly, the impact of treatment, in terms of manufacturing quality control, can be more clearly approximated, then equitable comparisons of defects identified under defined conditions, with those avoided under undefined conditions can be made, thereby providing a more exact determinate on how that alteration of operation impacts defects produced.

4.5.4 ATT and ATE Estimation

At the end of matched-pair analysis, two main measures, Average Treatment Effect on the Treated (ATT) and Average Treatment Effect (ATE), are to be calculated by the researchers. Calculating typical effect of treatment on units where treatment was actually received, the question to be asked is, what was the effect of the treatment in the treated? This step becomes especially useful in manufacturing, when the decision makers would like to know how a given change to their operations will affect the batches on which the change has been implemented. ATE on the contrary estimates the average effect of treatment of the entire population (both units taking the treatment and not taking the treatment) and thus provides a more generalised view of the general effectiveness of any particular treatment. The more relevant application of ATT in most industrial contexts is the fact that interventions typically aim at discrete batches, machines or processes and not the whole production process. As an example, the case study used to calculate ATTs and ATE to determine the effect of defects elimination which include broken stitches and bearing-hole problems on the overall quality of the products hence providing quantitative data to be used to back specific quality-enhancement efforts.

4.5.5 Scenario Simulation and Interpretation

Scenario simulation is probably one of the most practical applications of counterfactual analysis via propensity score matching. The technique involves manipulating certain treatment variables and observing similar projected changes in outcome of defect. In our example of jeans manufacturers, the respective investigators simulated the effects of the adjusted variable in the

sample consisting of high thread tension and discovered that the action resulted in the remarkable drop of the thread breakage and the buttonhole defects. These similitudes of conditions are as follows as: improper pocket marking and bent needles, where each of them was to test their potential impact on the severity of defects. The following interpretation of these simulations provides valuable input into managerial decision-making since the stakeholders will be able to foresee the consequences of interventions without practically carrying it on the production floor. This is what leads to significant cost saving and reduced downtime. However the interpretive phase requires very careful assessment, which is subject not only to statistical significance but also to a practical feasibility of operation. As an example, when the consumption of a certain defect causing factor has only a minor beneficial effect, the managers might consider such intervention not cost-effective enough. Such a simulation of scenarios balances the requirements of technical analysis with industrial reality in a particular way.

4.5.6 Decision-Making Support in Manufacturing

The main objective of combining the Process Sustainability Management (PSM) and counterfactual analysis is to support the decision-making potential in manufacturing. Traditional quality-control measures are inclined to rely on quality checks that take place at the end of the production process and they may take place after an accident has already happened. Mass comparison, on the contrary, allows the decision-makers to predict the processes and the conditions that need refinement as soon as possible before the deep-rooted problems with quality can be detected. In defining the best interventions, the manufacturing managers can thereby prioritize the possible interventions based on the potential magnitude of improvement, cost of the intervention and operational complexity. To take a concrete example, suppose that some MPV analysis suggests that re-setting thread tension would substantially decrease major defects and yet only modest investment would be necessary, then this measure would now be a no-brainer. In addition, counterfactual helps in resource allocation as it focuses on the improvement with the best effect. Instead of spreading resources to cover many minor problems, the management can focus where the highest returns are likely to be achieved. This approach is game-changing in small and medium-sized businesses where resources are always limited and decisions tend to be constrained by budgetary allocation and limited people as indicated in **Figure 22.**

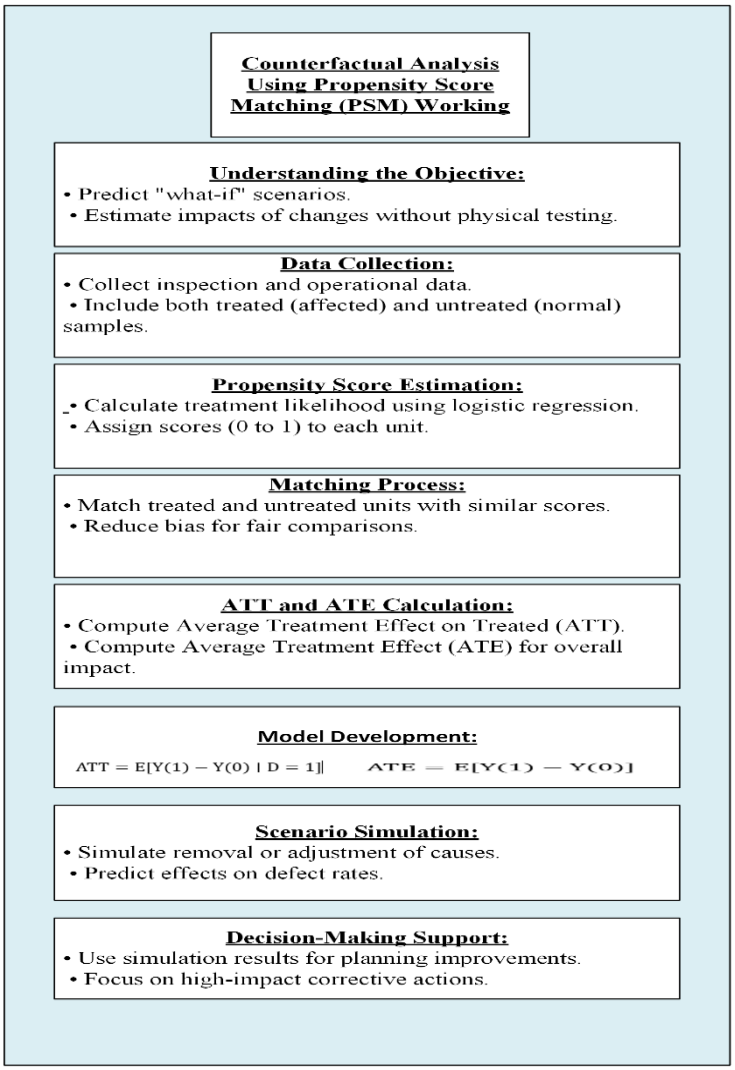


Figure 22 PSM Flowchart

Table 9 Advantages and Disadvantages of PSM

Advantages	**Disadvantages**
The method allows an estimate of a treatment effect of observational (non-experimental) data making it particularly applicable in the context of a manufacturing setting.	Effective use of the technique requires careful matching; any deviations might compromise the validity of the analyses and make it less credible.

Minimizes the bias in selection of the sample by balancing the covariates in the treated and untreated samples.	The statistical model cannot explain variable that may be unobserved and may affect treatment assignment fully.
With supports simulation of scenarios and predictive analysis, managers have a chance to see what their future might look like due to offered and potential changes.	The outcome of propensity-score is well conditioned on the model through which propensity scores are estimated and on the matching technique acquired.
Provided as a cheaper and more non-invasive alternative to costly and invasive real-life experiments.	The current analysis can overlook some units, which are poor matches and thus shrink the sampling size and it is possible to put limitations to the generalizability of the results.
The current model provides empirically based, measurable data that can lead to advancement of proactive quality improvement in manufacturing.	The proper execution of the procedure as well as the accurate interpretation of its findings requires a sound background in the statistical methodology.

4.6 Summary

The current chapter proposes a structured approach to the deployment of the Big Data Analysis in Quality Control (QC) of Small and Medium Enterprises (SMEs). It expresses the main steps data collection, cleaning, preprocessing, and further analysis, in order to provide tangible knowledge. Additional accessory advanced methods, in particular Multi-Layer Perceptron, the APROTRI Algorithm, and Propensity Score Matching, are being integrated to intensify the fault detection, redefine root-cause determination, and aid real causal inference. With such use of machine-learning technology and data analytics, SMEs can shift to the proactive quality management orientations, improving efficiency, increasing the product dependability, and improving managerial decision making. Finally, the approach described in the current article strengthens the long-term viability of business and predisposes the development of competitiveness within the industry.

Chapter 5. Results and Discussions

The given chapter contains detailed results of using the five-stage quality control model in the SMEs. The findings show that cutting-edge analytics and machine learning allow a major improvement of defect management.

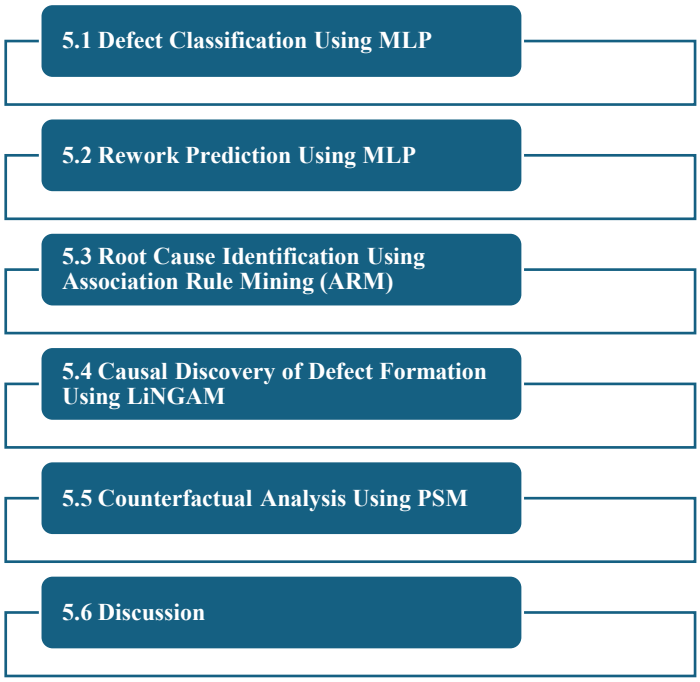


Figure 23 Flow chart Chapter 5

5.1 Defect Classification Using MLP

A Multilayer Perceptron (MLP) association of the study was prepared and trained to distinguish between the degrees of defects in a manufacturing database. In the classification process, the task was to identify the Minor, Moderate or Major defect utilizing the input attributes drawn out of a jeans manufacturing plant. The last model architecture had 3 hidden layers with 64 ,32 and 16 neurons respectively, where it used the Leaky ReLU as activation function. The Adam optimizer was used with a learning rate of 0.001 and a convergence was obtained after 50 epochs. During the training process, the training accuracy and validation accuracy were always higher than 99.13%, the model obtains final test accuracy is 99.03% which indicates high

predictive performance. Training loss was reduced steadily (0.999 to 0.004) whereas the validation loss was observed to follow the same pattern, reaching 0.005 and the two curves barely deviated. Such a unison of training and validation values is a sign of powerful generalization and demonstrates vividly that there is no overtraining procedure. That both loss curves meet around epoch 35 also assures the stability of the model and its effectiveness of learning the inherent trends in data. Detailed analysis of classification outcomes showed that most of the occasions, 53.9 percent were referred to Minor defects that can be characterized by surface-level scratches and cannot cause a lot of effort of correcting. Comparatively, the Moderate defects accounted 24.5 percent of the cases and usually contained functional types of defects that require moderate considerations of repair or correction. The rest 21.6 percent was termed as Major defects which are critical deformities that could either warrant much rework or complete rejection. The visual analysis of these proportions is provided in Figure 5, which allows the obtained picture of the defect landscape in the analyzed dataset to be understood. Successful use of the MLP model in this classification task proves its effectiveness as a tool of quality control at the industrial level as well. Automation of the defect severity classification will enable the manufacturers to implement tiered quality response systems which translate to increased allocation of human and material resources. This increases the efficiency of production besides aiding timely and informed decisions at the shop floor. In addition, this fact in combination with the high accuracy and stable performance of the model suggests that it can be deployed in other types of products and production lines, thus providing scalable advantages to date-driven quality management in production processes.

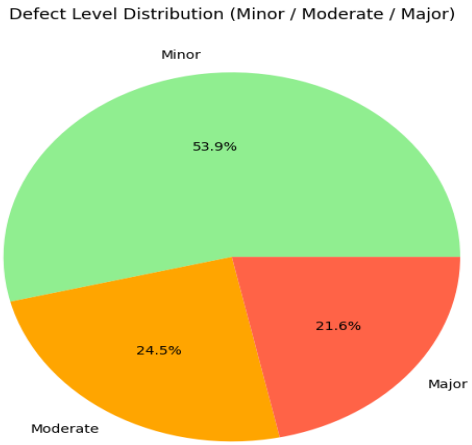


Figure 24 Pie Chart Analysis of Predicted Defect Severity Levels

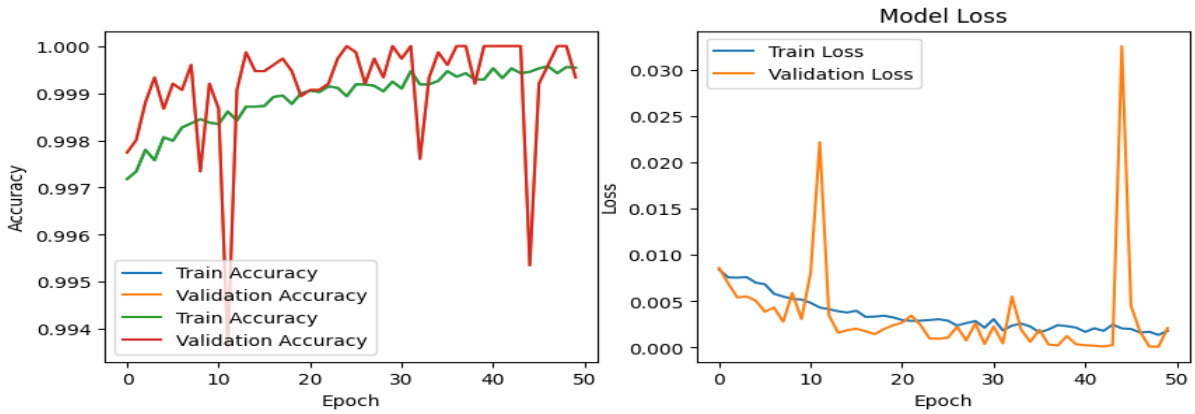


Figure 25 MLP Accuracy/Loss for Defect Classification

5.2 Rework Prediction Using MLP

In order to make an intelligent decision supportive to assistance in quality control, one made use of Multilayer Perceptron (MLP) to help in predicting whether the defect is to be rated under rework or under reject. The model has had a good performance with the final test accuracy of 99.75% and this reflects that the model will be able to distinguish the two types of defect successfully in a very precise production set up. It had good convergence in the training process of the model. The accuracy of the training increased progressively and eventually provided a value of 99.925 percent, and the accuracy of the validation started rising parallel with the training progression to achieve a value of 99.875 percent in the 50th epoch. The slight disparity of 0.05 percentage between the training and validation accuracy means that the model has a good fit in the unseen data and it does not overfit. Also, during the training process, training and validation loss were constantly decreasing. Training loss had a minimum value of 0.0018 and validation loss was between 0.0023 and 0.0040 indicating optimal stability in optimization. Figure 8 visualizes these trends as it illustrates the accuracy and dynamics of loss with time of the model. The confusion matrix below (Figure 7) was used to take a closer look at the classification results of the model. Findings indicate that that model was able to predict 16,530 rework instances and 2,275 reject instances and did not predict 19 instances as rework (false positives). Importantly, no false negatives were reported, which means a 100 percent recall could be achieved every true case of rework was recorded. This is imperative in production settings where the failure to detect reworkable elements may culminate in inconsequential

wastage or rejections. The structure of the model was lightweight (using 3 neurons on input layer, 16 and 8 neurons on the 2 hidden layers, and the sigmoid activation function on the last output layer) and could conduct binary classification. The invisible layers had the ReLU activation, adding value towards a good nonlinear features extraction. A well-tuned learning rate and schedule of epochs along with the Adam optimizer was used in training the network leading to a fast convergence and a small classification error. The rework prediction model has rigorous practical advantages. It can classify and discard more reliably and consistently in a controlled and safe environment than could be accomplished with manual types of inspections and thus helps to improve the results of the quality and results of efficiency. It is possible to use the system as an effective instrument in real-time quality control systems, and it would reduce inspection errors, speed up defect processing, and assist data-based quality assurance in a factory environment.

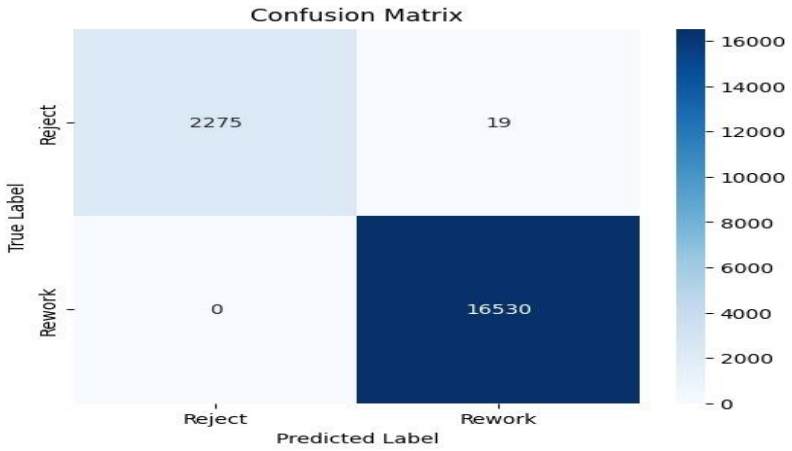


Figure 26 Confusion Matrix for Rework Prediction Model Performance

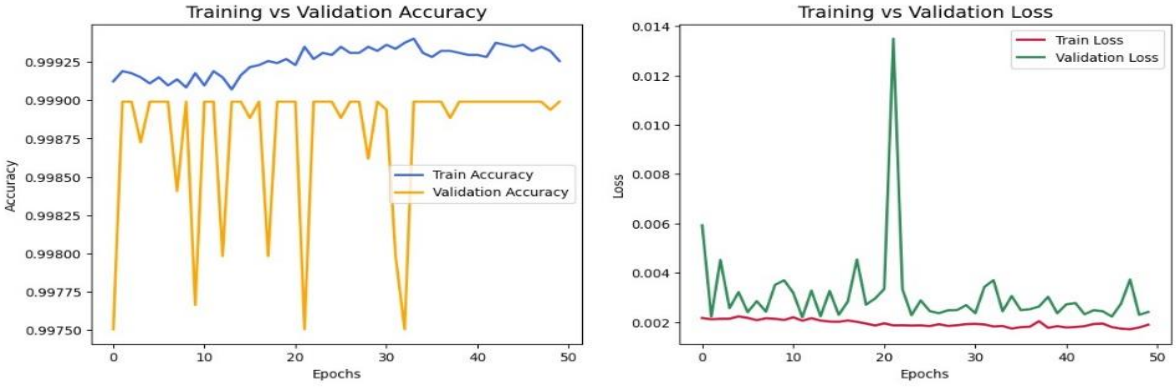


Figure 27 MLP Accuracy/Loss for Rework Classification

5.3 Root Cause Identification Using Association Rule Mining (ARM)

The analysis of association rule mining (ARM) carried out in this research employed lift standards to determine the degree of relationships between causes and outcomes of the defects in the manufacture of garments. The principles that came out of this approach enabled revealing of frequent and significant co-occurrences in quality related variables which gave an idea of the internal structure of the common problems in production. The individual rules are formed between an antecedent (one or more causes), and a consequent (the recorded defects) and the lift value acts as a guide to how firmly the two elements are related as more than by chance. When the lift value is above 1 it indicates there is useful interdependence among the variables and the antecedent is a significant contributor to the consequent. Various association rules were tested in the course of analysis to search the patterns of various combinations of defects. The rule of one of the most remarkable findings in them was as follows: “Broken Stiches, Buttons Hole Imperfections that had the biggest value of the lift 1.0181. This significant lift score demonstrates that, with the high likelihood, the co-occurrence of broken stitches and buttonholes defects is highly correlated with the threads tension problems. The orientation of this relationship was also established by its reverse rule which indicated that the relation is bidirectional. The other significant rule was related to improper marking associated with the process of pocket attachment which turned out to be constantly accompanied by visual inaccuracies and distortions in alignment such as color leaks and discrepancies in pockets. The significance of this connection also proves the necessity of procedural accuracy in avoiding various downstream flaws. Although an exhaustive list of rules (uncondensed) was also developed and checked during the analysis, it would be impracticable to provide all these rules as a visualization is a difficult task, and presentation would also be a problem. Therefore, only the top 10 rules as per lift value have been used to present the results. These are shown in Figure 9 which graphically summarizes most important and occur commonly persistent links between defects as witnessed in the data. The visualization also gives a clear picture of how remarkable discrete production problems can be traced to a similar in-common cause. Comprehensively, the findings of a lift-based analysis can be deployed towards the quality control initiative since this kind of analysis brings out patterns that would not be readily observable upon direct observation. The derived rules will be of some help to the manufacturer to design their inspection, training and process control strategies in a more selective way to reduce incidences

of defect clusters and in enhancing the general quality of the products.

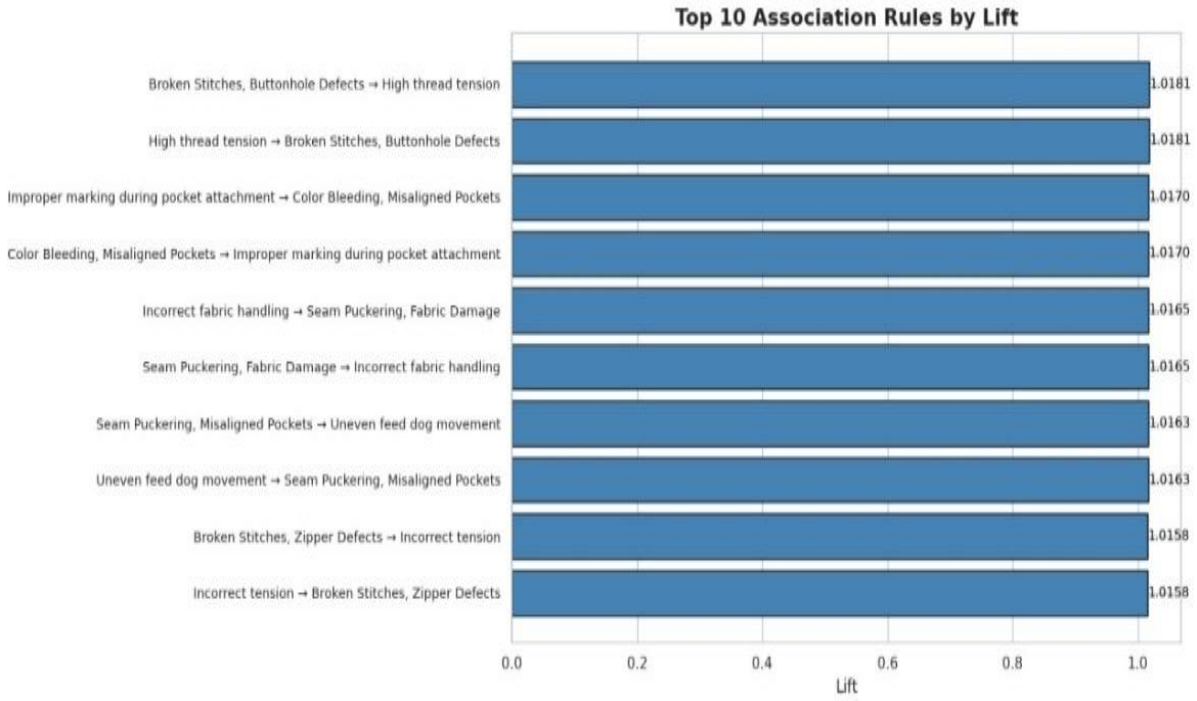


Figure 28 Top 10 Association Rules by Lift

5.4 Causal Discovery of Defect Formation Using LiNGAM

The LiNGAM-based causal discover can model production factors producing garment defects and demonstrate a complex network but which remains essentially clear to interpret. The scores of the likelihood of defects are higher, because of a number of factors that influence the propensity score variable. The seven defects that display the most damaging effect on the product are as follows: inappropriate work with fabric 0.34, bent or dull needles 0.31, excessive thread tension 0.34, excessive thread tension 0.26, inappropriate needle size 0.30, and inappropriate stitch type 0.27, sharp edges of machine elements 0.20 have emerged as the highest causes of defects in the products. In the evaluation, it can be shown that there are a number of intermediate causes each beginning with the incorrect thread tension leading towards high thread tension (0.18) and continuing to the improper stitch type (0.17). There are some variables that indicate negative relationships in the analysis. Defects include the seams puckering (-0.99), thread breakage (-0.37), broken stitches (-0.16) and those in zippers (-0.16) when poor-quality of thread is used their occurrence is lower. This indicates that the system has an early-stage so as to spot the low quality of thread in order to substitute it before

downstream issues arise. Also, certain defects will interact negatively in the sense that presence of one will decrease a certain number of defects as is in case of seam puckering whereby, it reduces three defects namely, fabric damage (0.45), color bleeding (0.43), and misaligned pockets (0.25) perhaps because of how the checkers make their adjustments during inspection **Figure 29**.

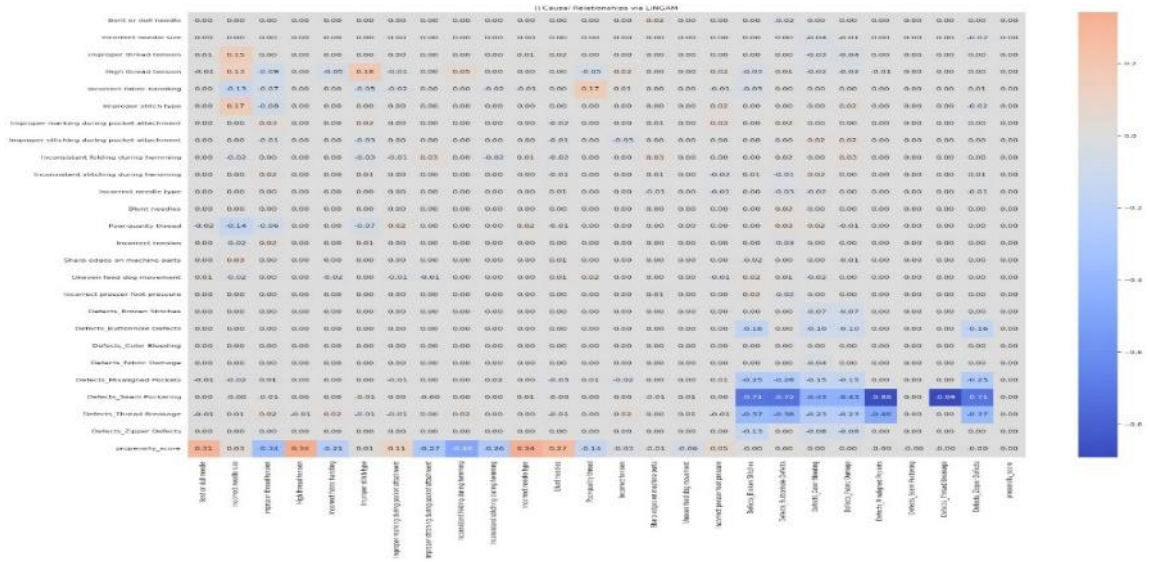


Figure 29 Causal Analysis Network

The details on the relationships between particular causes and individual defects are present in **Figure 30**. There is a positive correlation involving high thread tension and thread breakage (0.03) and damage on the fabric (0.03). With the combination of the wrong stitch type, one is more inclined to break the thread (0.02) along with damaging the fabric (0.03) and the wrong size of needle that causes both the issue of thread bleed (−0.04) and seam puckering (0.02). The incidence of poor thread quality is evidenced in negative correlations to the states of garment defects like seam puckering (-0.03) and zipper defects (-0.03) as faults in thread quality can be spotted and resolved prior to the products reaching the consumers.

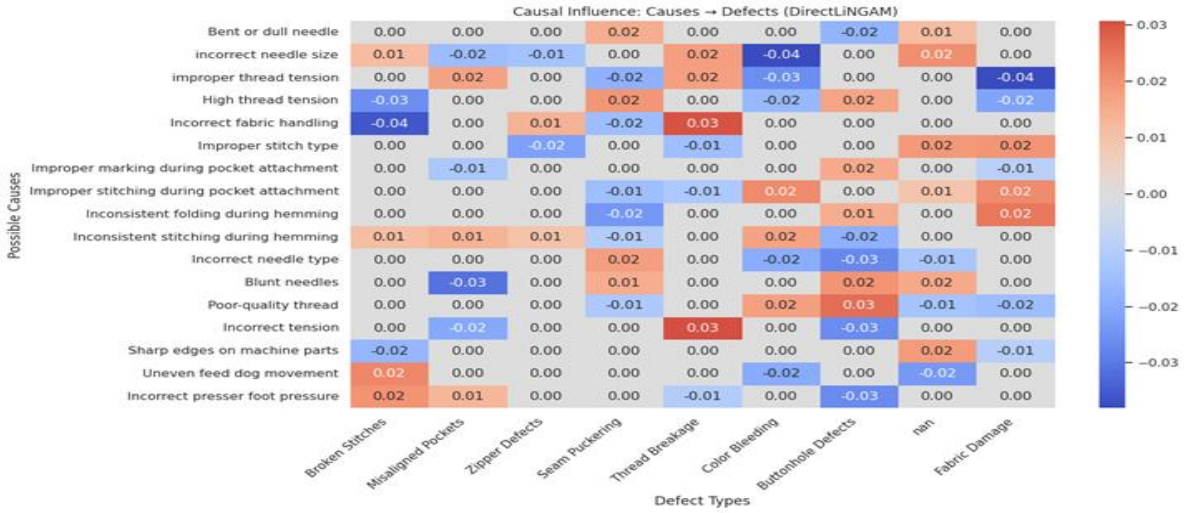


Figure 30 Relationships Between Causes and Defect

It has been demonstrated that there is the existence of an actionable causal structure that is followed by the development of a defect, which can be identified. **Figure 30** Connections between the Defined causes and the Types of Independence Defects Discovered by LiNGAM In the study an identifiable pathway of the causes resulting in formation of defects can be seen. Results of the study demonstrate that each needle state and wrong tension and handling introduced by an operator present quantifiable effects on defect types.

5.5 Counterfactual Analysis Using PSM

The impact of two causes of Misaligned Pockets was assessed with the help of the Average Treatment Effect on the Treated (ATT) analysis. When we removed “Incorrect fabric handling", "Improper marking during pocket attachment" and checking their effects on defect type Misaligned Pockets we got ATT value 0.06 for incorrect fabric showing that severity slightly rises when this factor exists which supports efforts to minimize defects by removing it. Pocket attachment marking performed properly resulted in an Average Treatment Effect on the Treated value of 0.00 which indicated no effect on defect severity as shown in **Figure 32**.

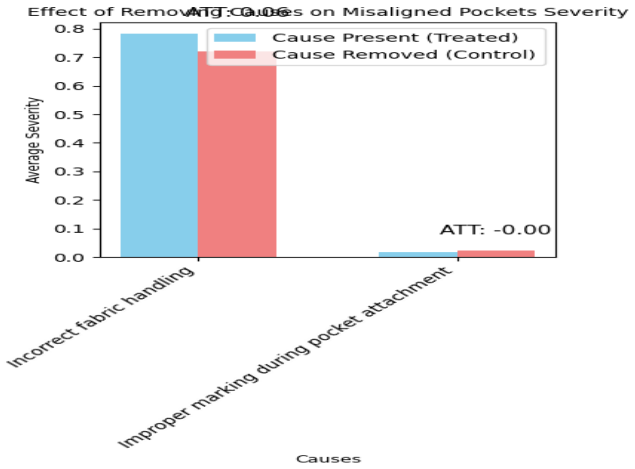


Figure 32 Impact of Fabric Handling & Marking on Pocket Alignment

The evaluation of buttonhole defect elimination occurred in the second scenario. When buttonhole defects existed, the average severity reached 0.70 but removing the defects from the matched control group reduced the average severity to 0.04. Research found that buttonhole defects reduction yielded an ATT value of 0.66 which shows strong evidence of minimizing defect magnitude as shown in **Figure 31**.

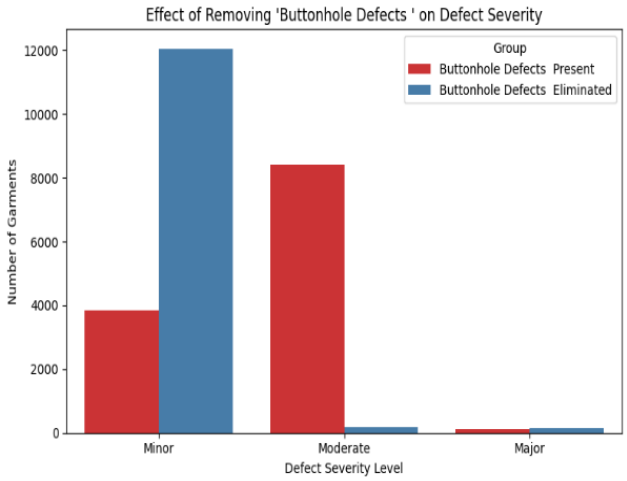


Figure 31 Severity Reduction After Buttonhole Defect Elimination

The buttonhole defects proved to be major contributors toward overall defect severity because the severity distribution showed increased minor defects and decreased moderate and major defects. The third scenario analyzed thread breakage as the main factor. The removal of this defect-causing factor triggered significant growth of minor defects, but severe defects fell dramatically as shown in **Figure 33**. The data shows thread breakage plays a leading role in producing high-severity defects and removing it will result in better quality garments. The

elimination of causes such as thread breakage and buttonhole defects reduce higher severity defects because their status moves toward the lower minor severity range. The insights can be used to work on their crucial product influences in their quality control strategy efforts towards the overall product quality.

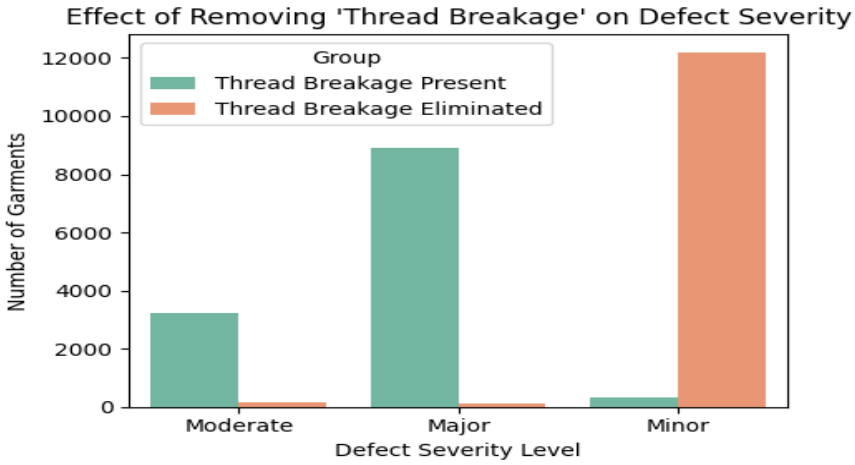


Figure 33 Severity Shift After Thread Breakage Correction

5.6 Discussion

The use of the five-stage analytics framework has led to obvious defect management and quality control. The paper showed that integrating machine learning, statistical analysis, and causal inference constructs a data-driven decision-making model to be used in manufacturing. The models designed were precise, had extensive generalization, and practical value and most easily permitted automated defect classification and prediction. Association analysis was used to reveal the instances and the concealed connections between causes and defects, and causal discovery was used to encapsulate the many-sidedness of the role played by definite factors with regard to the formation of defects. Counterfactual simulations also demonstrated the practical effect of improvement efforts in targeted interventions and provided specific directions to quality efforts. On the whole, the results confirm the conclusion that advanced analytics is a considerable instrument of efficiency increase and waste minimization in the context of the industrial production.

Chapter 6: Environment and Sustainability

It is an important chapter that evaluates critically the environmental and economic impacts of adoption of data-driven quality management system in Small and Medium Enterprises (SMEs). On the basis of superior procedures- such as- machine learning, causal inference, counterfactual analysis, the chapter is used to depict that the superiority of proposed framework not just raise operational perfection of extraordinary folds, but to stimulate a enterprise as far as sustainability and corporate responsibility is concerned. The shift of SMEs to predictive quality systems is capable of majorly preventing environmental degradation, optimize resources usage and improve long-run economic sustainability.



Figure 34 Flow chart Chapter 6

6.1 Environmental Impact of Data-Driven Quality Management

The impact it has on the environment by traditional quality control systems within SMEs is contemptibly ignored. Such systems with slow identification of defects and high volume rework lead to excessive wastes, poor energy consumption, and increased carbon footprint. The

suggested data-driven model will be a radical alternative. With defect classification early on, predictive maintenance, and the best process control, the system can proactively reduce wastage of materials, decreased energy use and cut down vacuous production iterations. This has a direct effect of quantifiable lower impact on the environment.

Table 10 Environmental Impact: Data-Driven vs. Conventional Quality

Environmental Indicator	**Data-Driven Approach**	**Conventional Approach**
Waste Generation	Alternatively, production of defects early means less scrap and reprocessing.	Increased scrap rates because of low levels of detection and remedial sections.
Energy Efficiency	Simplified activities minimize power consumption on per production unit basis.	Wearing down and repairs increase the rate of consumption of energy.
Resource Utilization	AI guarantees accuracy in matching the input, and curbs overproduction.	Raw materials are usually overused due to manual estimation.
Carbon Emissions	Reduces as a result of reduced number of rework cycles and decreased production time.	Increased as a result of ineffective work processes and replication manufacturing

6.2 Economic Benefits of Sustainable Quality Management

There are deep economic implications of the fact that the use of advanced analytics is becoming integrated into manufacturing processes. Investment on the smart systems, IoT sensors and analytics infrastructure can be massive but the payoff in the reduced number of defects, increased throughput, reduced rework and rejection expense offers a high return on the investment (ROI).

Table 11 Economic Comparison: Smart vs. Traditional Quality

Economic Metric	Data-Driven Quality Management	Traditional Quality Control
Capital Investment	It needs the initial investment in the digital infrastructure.	Low initial cost followed by lowering of operation.
Labor Efficiency	Automation helps minimize on manual inspection.	Expensive in labor in checking out the repetitive matters as well as resolution of matters.
Scrap & Rework Costs	Reductions caused by the prevention of issues.	Costly because of the correctives made once defects have been made.
Long-Term Competitiveness	Improves a market position in terms of innovation and efficiency.	Reduction of returns and falling competitiveness with time.

The framework will give firms a leg up by reducing scrap recycling and repair and enable re-investment in technology and people, so that even in a turbulent industrial environment, the company will be financially stable.

6.3 Environmental Impact Assessment

An orderly environmental impact analysis confirms the sustainability level of the proposed system. Environmental indicators experienced were compared and contrasted in order to assess the result during both conventional and data-driven regimes.

Table 12 Sustainability Metrics: Proposed vs. Conventional Practice

Sustainability Metric	Proposed Framework	Conventional Practice
Carbon Emissions	A lot less due to optimized cycles and a lesser number of defects.	High because of too much rework and too inefficient operations.
Energy Consumption	Very expensive due to over rework and poor operations.	High due to machine break down and bad scheduling.

Material Waste	Managed through the predictive analytic and error spotting in advance.	Important since it did not offer predictive nuggets.
Chemical & Water Usage	Reduced through process optimisation and through mitigation of the root cause.	Non-controlled use on defect-intensive periods and during the rework tasks.

These findings show the direct support provided by the AI-driven quality control toward the green manufacturing and cleaner production philosophies.

6.4 Contribution to Sustainable Development Goals (SDGs)

It is a highly viable technical (as well as ethical) progression in contemporary manufacturing which complies well with the United Nations Sustainable Development Goals (SDGs). Certain SDGs which are being tackled are:

- **SDG 8 – Decent Work and Economic Growth**: Automation and upskilling will be promoted through this system where innovation-based employment and productivity improvement will take place.
- **SDG 9 – Industry, Innovation, and Infrastructure**: They promote the use of smart infrastructure and an industry that is innovation oriented within the SMEs.
- **SDG 12 – Responsible Consumption and Production**: Enables the minimization of wastes, efficient use of material and sound process design.
- **SDG 13 – Climate Action**: Is environment friendly as it lowers the emission of greenhouse gases through more efficiency and less exploitation of unnecessary resources.

The framework enhances the role of SMEs to the active player in sustainable industrial change through its contribution towards these global goals.

6.5 Trade-offs and Barriers to Implementation

Although it seems to have obvious merits, the process of implementing a data-guided quality system implies a number of trade-offs which should be carefully managed:

- **Capital Barriers:** Start up hardware, software and training costs can be a barrier to small scale enterprises and micro enterprises.
- **Technical Literacy:** Majority of SMEs do not have staff that is skilled at the data science, analytics, and AI based tools.
- **Level of Complexity:** Deployment can be slowed due to the incompatibility of the current legacy systems.
- **Phase Deployment:** The gradual and start with pilot projects and become larger.
- **Training Programs:** Collaborate with educational institutions and official organizations to train the employees.
- **Policy Support:** To gain advantage of green technology, use incentives to adopt green technology like tax relief or subsidy advantage.
- **Hybrid Architecture:** Deploy modular structures which are capable of being integrated with both the analog and the digital infrastructure.

The methods aid in striking the trade-offs and maximize the value that can be exploited in the upgrade of technologies.

6.6 Strategic Recommendations for Sustainable Manufacturing

To help SMEs on the way to eco-efficient production, the provided recommendations are suggested:

1. **Embed Predictive Maintenance:** Avoid the occurrence of unexpected failures of the equipment and increase the energy consumption efficiency applying ML algorithms.
2. **Maximize Process Intelligence**: Assess the root cause logic as well as counterfact logic of eliminating the inefficiencies that are in the system.
3. **Adopt Renewable Energy Integration**: Utilize green sources of energy (e.g. Solar, wind), combined with AI enabled optimization resources.
4. **Establish Ecosystem Collaboration**: The co-creation of design among suppliers, logistic providers, and customers by power of a circular economy.

Combined with each other, these practices can give SMEs power to therefore be eco-efficient and regulatory compliant as well as relevant in the global market.

6.7 Chapter Summary

This chapter presents a net review of environmental and economic advantage of a solitary driven quality management technique adjusted to SMEs. The model can help mitigate wastage, energy consumption and emissions, enhance the quality of the products and economic sustainability. In addition, it goes hand-in-hand with the SDGs which strengthens its contribution to and global sustainability agendas. The costs of implementing the barriers notwithstanding, in terms of its long-term strategic advantages, such as cost savings, increased competitiveness and environmental responsibility, the transformation is not only achievable, but a necessity. The synergy of AI, analytics, and sustainability thinking is a new paradigm of manufacturing embedded with a power to encourage SMEs to succeed within the Industry 4.0 and the era of sustainable development.

Chapter 7: Conclusion and Recommendations

The implementation of a data-centric quality management infrastructure whereby the predictive and inferential functionality forms the backbone of the infrastructure has become a game-changer in the area of quality management and quality operational excellence optimization in Small and Medium Enterprises (SMEs). The current study reports the synergistic workings of Multi-Layer Perceptron (MLP), Association Rule Mining (ARM), and causal inference based on LiNGAM, which completes a set of measurable improvements on fault prevention, rework and root-cause studies as **Figure 35**.

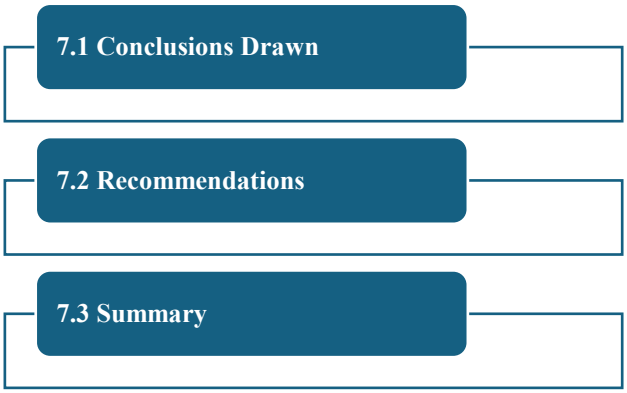


Figure 35 Flow chart Chapter 7

The most promising part of the initiative is its systematic response to the chronic efficiency of complaints as it moves further towards an organically anticipatory structure, creating a manufacturing ecosystem that looks to the future and makes future-specific conclusions. The consequences are far-reaching: the derived outcome is harmonious with the global objectives that are outlined by the United Nations Sustainable Development Goals (SDGs)

- SDG 8: Decent Work and Economic Growth
- SDG 9: Industry, Innovation and Infrastructure
- Goal 12: Responsible consumption and production
- SDG 13: Climate Action

7.1 Conclusions Drawn

The approach proposed within the framework of the current project was sequential, utilizing data collection, data preprocessing, algorithmic modeling, root-cause identification, and

counterfactual simulation strategies. The project revealed convincing findings and confirmed a number of strategic decisions

Table 13 Conclusions from Project Completion

Sr. #	Conclusion	Description
1	Product Quality Classification	The used classification was able to precisely distinguish the different severity levels of both classes of issues of the products and thus led to consistency and substantial simplification of speed and objectivity, significantly more than what was offered by the manual assessments.
2	Decision Support for Rework	The examined decision-making framework was demonstrated to be highly reliable in identifying whether or not the items needed the additional efforts of being reworked and whether they should be discarded in order to support the optimal efficiency of the resourcing activities and enabling superior resource management terms of waste reduction.
3	Pattern Recognition in Operations	A thorough analysis of past data has revealed the existence of rock-solid and hitherto latent linkages of production environments and frequent complications thus providing a complete insight into the quality challenges of various processes.
4	Identification of Root Causes	A stringent causal inquiry has identified a few process-based factors as key forces along product quality. Such considerations make it possible to rank the areas that need remediation accurately and highlight unexpected effects of current operational processes.
5	Scenario-Based Quality Forecasting	Simulation outcomes show that eliminating the specified influential variables considerably enhances quality results, which serves as additional evidence of the need in proactive planning and the strategic orientation of specific process improvements.

7.2 Recommendations

To maximize the success of the current framework and increase its industrial and environmental impact the following is proposed:

1. **Connection of the Internet of Things and Real-Time Observation**

   As sensors in the form of the Internet-of-Things technology are introduced along manufacturing lines, it is now possible to track defects in real time, use early warning signals and control automatically. This enhances accuracy of managerial decision making as well as responsiveness of the system.

2. **Advanced AI Model Adoption**

   Any further investigation should employ sophisticated deep learning architectures, in particular, Convolutional Neural Networks (CNNs) and Long Short-term memory (LSTM) models to recognize detailed defect patterns, as well as learn dynamic production behavior.

3. **Capacity Building and Employee Training**

   The framework is to be complemented by custom training modules based on ARM and LiNGAM on the formation of the reasons of defects, increasing the level of skills and working conditions of workers. This is in support of Sustainable Development Goal 8.

4. **Cross-Sector Scalability**

   One can recommend extension of this framework to other industries, other than textile industry, as textile, automotive, electronics, and pharmaceutical industries, specializing defects and production parameters to suit the industry.

5. **Platforms to join the Cloud-Based Analytics**

   Interactive dashboards and platform in the clouds must be created to show the key performance indicators (KPIs), real-time process analytics, and defect heatmaps that in turn will help make more knowledgeable managerial decisions.

6. **Integration of Sustainability Policy**

   They should be enabled to coordinate their operations of the small- and medium-sized enterprises with the regulation of sustainability by collaborating with governmental authorities and industry associations and also by encouraging environmentally friendly

technologies, which is to be fulfilled under the Sustainable Development Goals 12 and 13.

7. **Modeling of Environmental Impact**

   Its framework must be extended to include carbon footprint estimations, energy consumption indicators and waste-heat recovery models, and directly align with climate mitigation targets of Sustainable Development Goal 13.

7.3 Summary

This chapter has produced the evidences of finding and strategic implications of the research project. By adopting the multi-stage approach, which is sequencing of data preprocessing followed by defect classification (MLP), pattern discovery (ARM), causal analysis (LiNGAM), and counterfactual evaluation (PSM) steps, the proposed framework provided drastic gains in the area of operation efficiency and sustainability.

Table 14 Framework Applications and SDG Alignment

Application Area	**Impact**	**SDG Alignment**
Defect Prevention	Better product quality with greatly lowered rates of defects using the data-based prediction methods.	SDG 9, SDG 12
Resource Optimization	Reduced material and energy as well as the throughput of the process	SDG 12, SDG 13
Environmental Stewardship	Promoted environment friendly production through less rework and emission	SDG 13
Workforce Development	Improving the effect and satisfaction in jobs and training	SDG 8
Digital Innovation	Incorporated the usage of AI and ML with SMEs.	SDG 9
Operational Sustainability	Due to the enhanced productivity and affordability of the manufacturing process, there is also enhanced resilience and affordability of production.	SDG 8, SDG 12

7.4 Final Reflection: A Roadmap for a Sustainable Industrial Future

Besides achieving project technical milestones, it established a feasible blueprint of an intelligent, sustainable, and socially-safe operation of industrial production, which can be scaled up to national platforms and beyond. Giving SMEs with some predictive models and analytical tools, the framework assists to shift them towards:

- **Eco-friendly production practices (SDG 12 & 13)**
- **Technological innovation and infrastructure modernization (SDG 9)**
- **Stable employment, improved work environments, and skill development (SDG 8)**

Industrial quality management requires data-driven transformation, and the project proves this idea. In the era of climate urgency and Industry 4.0, such innovations are not an option anymore because they will allow countries to be competitive in the long term, as well as environmentally and socially prosperous.

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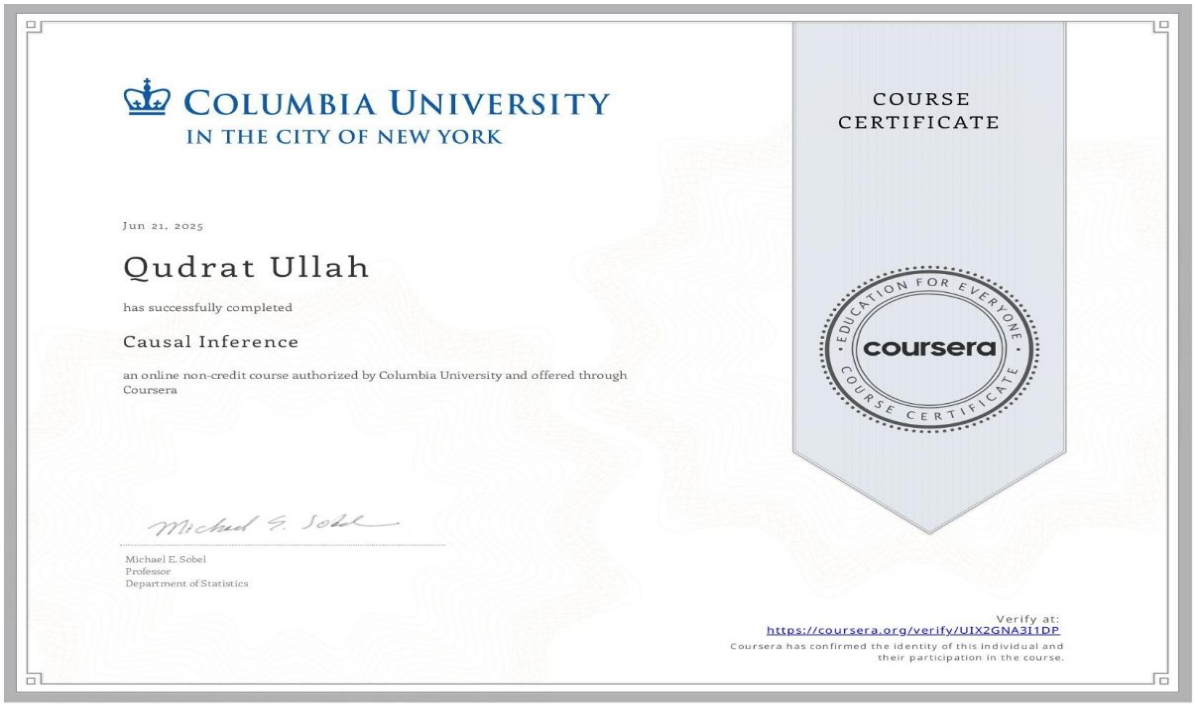
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Appendices

Appendix – A

International Certifications



COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

COURSE CERTIFICATE

Jun 21, 2025

Qudrat Ullah

has successfully completed

Causal Inference

an online non-credit course authorized by Columbia University and offered through Coursera

Michael E. Sobel
Professor
Department of Statistics

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Certificate no: UC-8fd42db1-4c48-48b3-947d-2426a04fbded
Certificate url: ude.my/UC-8fd42db1-4c48-48b3-947d-2426a04fbded
Reference Number: 0004

CERTIFICATE OF COMPLETION

2025 Complete Python Bootcamp from Zero to Hero in Python

Instructors **Toppers Bootcamp**

Qudrat Ullah

Date **May 3, 2025**
Length **11 total hours**


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Certificate
u6242697
We certify that Qudrat Ullah
has completed the following course on our learning platform:
Machine Learning tutorial by Great Learning
10 hours 20 minutes - Completed on: 04/30/2025
Qudrat Ullah
Student Signature
Qudrat Ullah
Adrian Medeiros Dantas
Executive Director - Cursa
Point the camera to
verify authenticity

COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

COURSE CERTIFICATE

Jun 21, 2025

Muhammad Ali Raza

has successfully completed

Causal Inference

an online non-credit course authorized by Columbia University and offered through Coursera

Michael E. Sobel
Professor
Department of Statistics

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https://coursera.org/verify/AGUJKV53UP9V
Coursera has confirmed the identity of this individual and their participation in the course.

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COURSE CERTIFICATE

Feb 6, 2025

Muhammad Ali Raza

has successfully completed

A Crash Course in Causality: Inferring Causal Effects from Observational Data

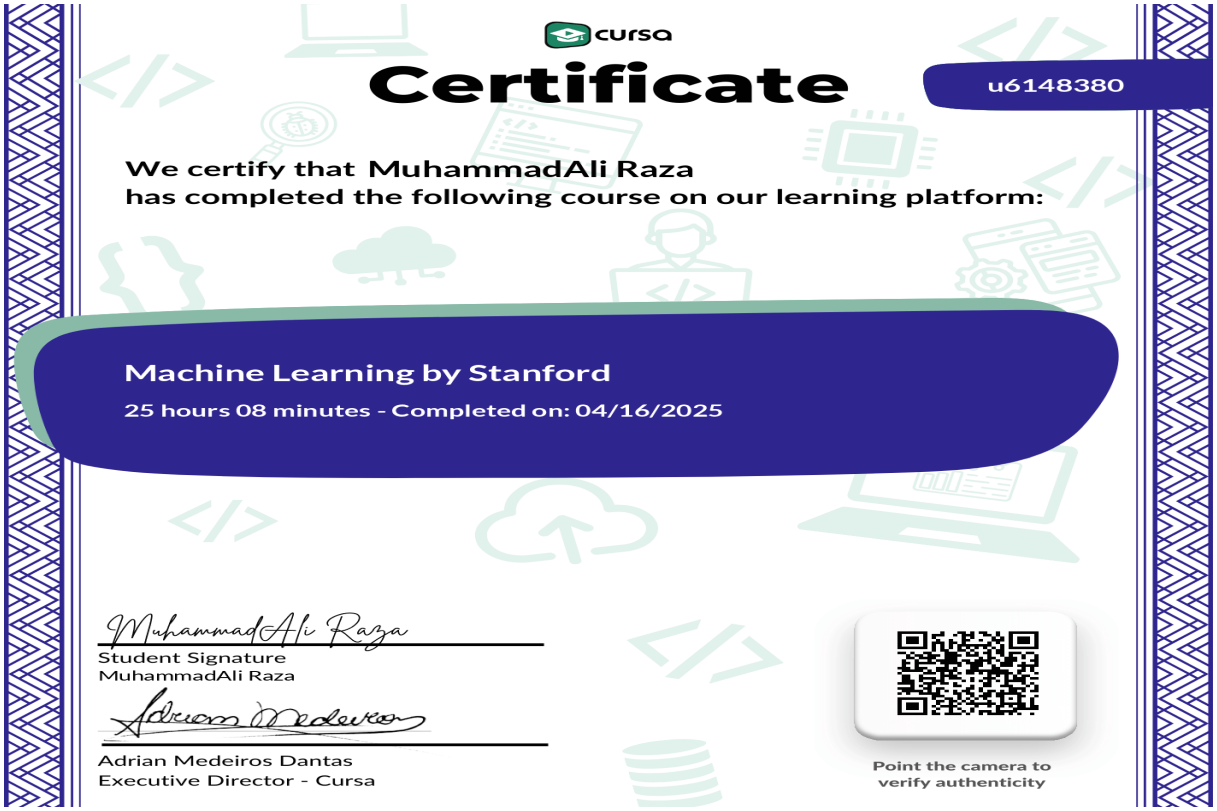
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24th Aug 2024
Krishna Kumar
CEO, Simplilearn


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Certificate
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We certify that MuhammadAli Raza
has completed the following course on our learning platform:
Machine Learning by Stanford
25 hours 08 minutes - Completed on: 04/16/2025
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Student Signature
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Appendix – B

Table 15 Attributes of Complex Engineering Problem (CEP) – Mapped with Project

Attributes	Description
Complex Problems	The project addresses the complex quality control problems of SMEs in the realms of the ML-driven defect prediction, defect classification, and root cause analysis.
Preamble	The motivation under the work is to minimize rework, optimum utilization of the resources, and transition in reactive quality systems to a preventive system.
Range of Conflicting Requirements	Is high-accuracy, low-latency, industry-scalable and usable by operators.
Depth of Analysis	Necessary performing and adjusting supervised and causal ML models and Counterfactual analysis.
Depth of Knowledge Required	Use the knowledge of machine learning, industrial engineering, manufacturing systems, and analytics.
Familiarity of Issues	Problems relating to high level of defects, the lack of traceability of root cause and non efficient manual inspection were solved.
Extent of Stakeholder Involvement	The main stakeholders participating in it include production managers, quality engineers, machine operators, and owners of SMEs.

Appendix – C

Table 16 PLOs Mapping

PLOs	**Ch#1**	**Ch#2**	**Ch#3**	**Ch#4**	**Ch#5**	**Ch#6**	**Ch#7**
PLO1: Fundamental engineering concepts	✓						
PLO2: Research and analysis of complex problems		✓					
PLO3: Design solutions			✓				
PLO4: Investigation of complex problems				✓			
PLO5: Use of appropriate tools and techniques					✓		
PLO6: Consideration of societal implications						✓	
PLO7: Consideration of environmental implications						✓	
PLO8: Consideration of ethical implications							✓
PLO9: Teamwork					✓		
PLO10: Effective communication							✓
PLO11: Project management	✓						
PLO12: Lifelong learning							✓

Appendix – D

Table 17 Sustainable Development Goals (SDGs) Mapping

Sustainable Development Goals (SDGs)	**Relevance to the Project**
SDG 8: Decent Work and Economic Growth	The framework will enable the SMEs to increase productivity and minimize costs, which translate into better working conditions, employment security and production effectiveness.
SDG 9: Industry, Innovation, and Infrastructure	With the usage of AI and ML in quality management, the project will help to modernize the industry and implement intelligent technologies of manufacturing.
SDG 12: Responsible Consumption and Production	Eliminates wastages of materials, minimizes repetition and promotes efficiency of resources, which is responsible manufacturing.
SDG 13: Climate Action	Indirectly, the goal of minimizing wastage of resources and energy inefficiency is linked to the reduction of degradation of the environment and supersonic change.